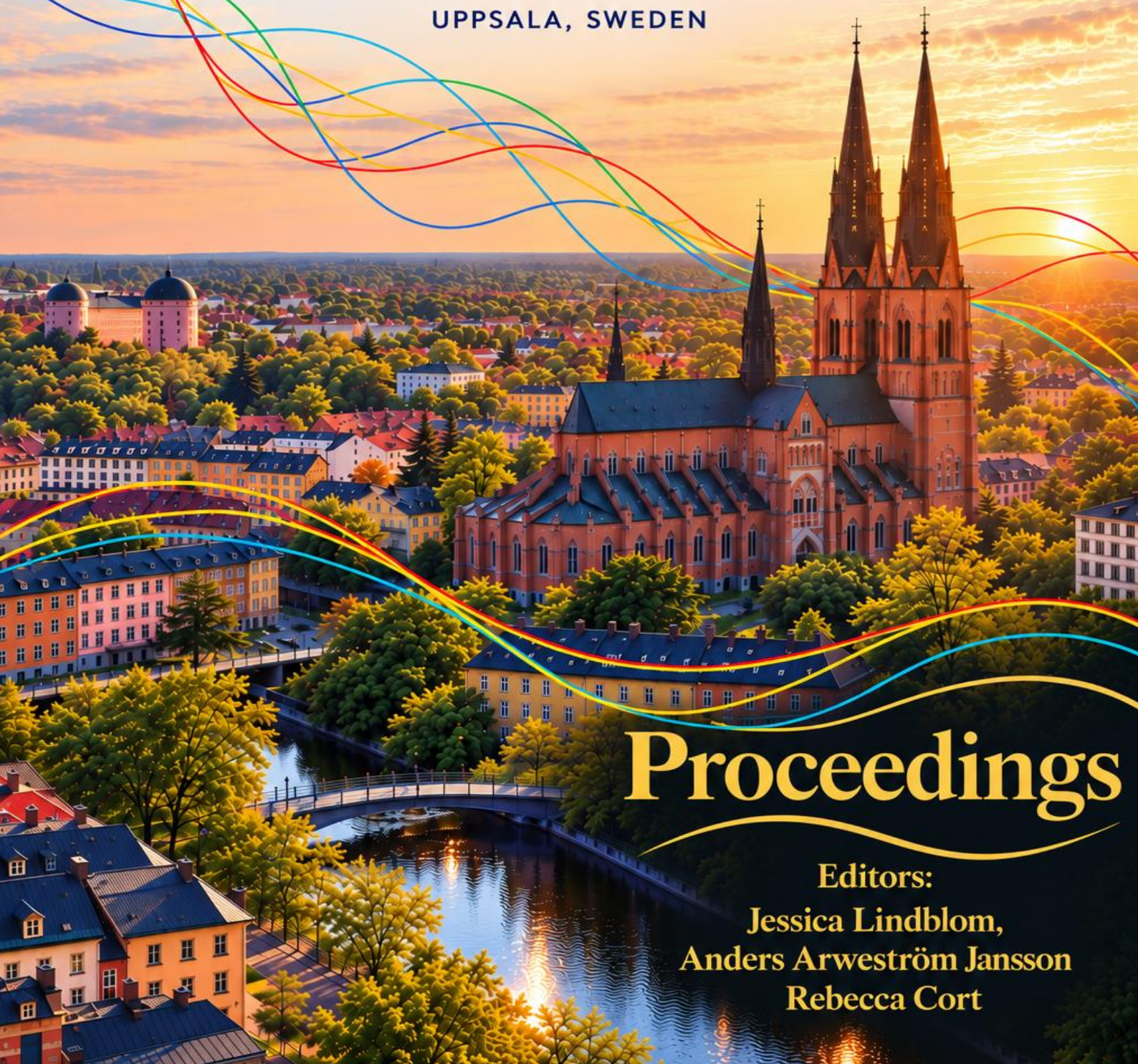
SweCog
SWEDISH COGNITIVE SCIENCE SOCIETY

UPPSALA UNIVERSITET

SweCog 2026

CONFERENCE | 26 – 28 OCTOBER 2026

UPPSALA, SWEDEN



Proceedings

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DOI:10.5281/zenodo.23666638
Published by Uppsala University and DiVA
Digitally available at: https://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-602242

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Preface

Welcome to SweCog 2026 in Uppsala!

We are delighted to welcome you to SweCog 2026, *the 21st Annual Conference of the Swedish Cognitive Science Society*, held on 26-28 October at the Ångström Laboratory, Uppsala University.

Uppsala is a city with a long academic tradition and home to two universities: Uppsala University and the Swedish University of Agricultural Sciences (SLU). Together, the two universities contribute to a rich and diverse research environment spanning many disciplines. With its historic surroundings, vibrant student life, and contemporary research environments, we hope Uppsala provides an inspiring setting for three days of cognitive science.

This year's programme illustrates the breadth that characterises cognitive science. Across four keynote presentations, seven oral sessions, poster sessions, and a workshop, SweCog 2026 brings together research on AI and technology, linguistics, learning, interaction, memory, perception, social cognition, human-robot interaction, as well as animal cognition and welfare. The programme spans several disciplines, theoretical perspectives, and methodological approaches, while sharing an interest in understanding cognition in its many forms and contexts.

We hope the conference will provide opportunities not only to present and discuss research, but also to meet across disciplinary boundaries, exchange perspectives, and establish new connections. We also hope you will find some time to experience Uppsala beyond Ångström – from the Cathedral, Uppsala Castle, and historic university quarters to the Linnaean Garden and Old Uppsala.

We warmly thank all keynote speakers, presenters, reviewers, session chairs, volunteers, and participants whose contributions have made SweCog 2026 possible. We also gratefully acknowledge the financial support from the Trustworthy AI Arena at Uppsala University. We would also like to extend our sincere thanks to Lisa Wästberg at Destination Uppsala for her kind support and assistance in the preparations for SweCog 2026. We greatly appreciate her help.

We wish you inspiring conference days filled with stimulating discussions, new ideas, and new connections!

A warm welcome to Uppsala and SweCog 2026!

Jessica Lindblom, Conference and Programme Chair,
on behalf of the Conference and Programme Organisers

Uppsala 23 October, 2026

Organisers, Chairs and Reviewers of SweCog2026

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Katja Valli, University of Skövde

Keynote Speakers

Returning to the Elusive Concept of Representative Design

by

Peter Juslin



Abstract
Given the strong Brunswikian tradition at the Department of Psychology at Uppsala University, I take this opportunity to return to a discussion, evaluation, and elaboration of the elusive Brunswikian concept of Representative Design. The logic of the concept, as such, is often endorsed in principle, but its application often becomes lost in endless and less fruitful discussions on how to define and operationalize people's "everyday environment". I will briefly review applications of representative design in our previous research, as well as as in recent study on the transitivity of preferences. After discussing different interpretations of the concept, I will argue in favor of a more specific and technical definition of the concept of representative design than often assumed, which better meets the challenges of operationally defining the natural sampling population and with a (moderately) revised rationale.

Peter Juslin is Professor of Cognitive Psychology at the Department of Psychology, Uppsala University. His research focuses on human judgment and decision-making, including how people integrate information, make judgments under uncertainty, and combine intuitive and analytical forms of reasoning. His research has addressed topics such as overconfidence, probability judgment, multiple-cue judgment, and the idea of humans as "naïve intuitive statisticians.". Peter Juslin is a member of the Royal Swedish Academy of Sciences.

https://www.uu.se/en/contact-and-organisation/staff?query=XX3971

What Would Evolution Build? An Evolutionary Perspective on Associative Memory

by

Mats Lind

Abstract
Associative memory refers to the retrieval of stored information from partial or related cues, rather than from an explicit index. It is therefore a useful meeting point between psychological theories of memory and computational models such as exemplar-based representations, Hopfield networks, and linear associative memories.

Many computational approaches to associative memory involve important trade-offs. Classical Hopfield networks, for example, are attractor-based models of content-addressable retrieval, but their storage capacity is limited and they can develop spurious states. Similarly, linear associative memories offer mathematical simplicity and potential biological relevance, but their performance can be constrained by interference, representational overlap and capacity limitations. However, recent developments, including modern Hopfield networks and new approaches to ensembles of optimal linear associative memory banks, suggest that some or most of these limitations can be substantially reduced.

If computational limitations alone do not determine which model is most appropriate, what other principles should guide model selection in psychological research? In this talk, I will argue that evolutionary considerations provide one useful source of constraints. Different memory architectures imply different trade-offs among storage capacity, robustness, generalization, interference, and flexibility. Considering the adaptive problems that ancestral organisms had to solve over evolutionary time may therefore help us identify a small set of computational architectures that are plausible candidates for modeling memory in humans and other animals.

Mats Lind is Professor Emeritus in Human-Computer Interaction at the Department of Information Technology, Uppsala University. His research has focused on human perception and cognition, particularly visual perception, spatial cognition, and the interaction between perceptual processes and technology. His work spans cognitive science and human–machine interaction, with a longstanding interest in how humans perceive, interpret, and interact with their environment.

https://www.uu.se/en/contact-and-organisation/staff?query=XX2868

Cognition and Emotion in Animals: Implications for Animal Welfare

by

Linda Keeling and Else Verbeek





Abstract
Animal welfare concerns how sentient animals experience their lives. In this presentation we outline several existing methods as well as new trends in the study of cognition and emotion in animals to assess their welfare. We present examples from our own work investigating cognitive judgement bias, attention bias and body language in domesticated species, together with studies on other species. We also investigate whether opportunities to make meaningful choices and exercise control could help young animals develop capacities that support good welfare throughout life. Finally, we consider how increasing interest in positive animal welfare in combination with a greater appreciation for the cognitive abilities of animals might mean for our relationships to animals in the future.

Linda Keeling is Professor of Animal Welfare at the Department of Applied Animal Science and Welfare, Swedish University of Agricultural Sciences (SLU) Uppsala, and an internationally leading researcher in animal behaviour and welfare science. Her research explores fundamental questions about animal cognition, emotion, motivation, and social behaviour, with a particular interest in how we can understand and assess animals' positive emotional experiences and how early-life experiences shape later behaviour. Her work bridges fundamental research on animal minds and behaviour with major societal questions concerning animal welfare and sustainable development. She has published extensively in the field and leads the Centre of Excellence in Animal Welfare Science, an interdisciplinary network bringing together around 100 researchers. Beyond her academic research, Keeling has played an influential role in translating animal welfare science into policy. She is an independent expert member of the EU Platform on Animal Welfare, has served on the

European Food Safety Authority's Animal Health and Animal Welfare Panel, and has chaired the Biology Committee of the Royal Swedish Academy of Sciences. She is a member of the Royal Swedish Academy of Agriculture and Forestry and an Honorary Doctor at Linköping University.

https://research.slu.se/en/persons/linda-keeling/

Else Verbeek is Associate Professor in Animal Welfare and Ethology at the Department of Applied Animal Science and Welfare, the Swedish University of Agricultural Sciences (SLU), Uppsala. Her research seeks to understand why animals behave the way they do and how insights into behaviour, cognition and emotion can be used to improve animal welfare and care. Her research spans animal behaviour and welfare, with particular interests in cognition and emotion, problem behaviours, stress physiology, neuroendocrinology, and the gut-brain axis in dogs. Her current work includes research on cognition and emotion in dogs, aimed at developing better ways of understanding and addressing behavioural problems such as separation anxiety. She also investigates the relationship between gut microbiota, behaviour and cognition in pigs, including how prebiotic and probiotic interventions may influence behaviour and quality of life. With more than 18 years of research experience, Verbeek combines behavioural, cognitive and physiological perspectives to understand animals' emotional experiences and responses to their environments. Alongside her research and teaching at Bachelor's, Master's and PhD level, she is committed to translating scientific knowledge into evidence-based practice. Through the education platform Amazing Animal Minds, she brings current research on dog behaviour, dog cognition, stress and emotion to professionals working with dogs and their welfare.

https://internt.slu.se/en/cv-originals/else-verbeek/

Designing Robots, Designing Society

by

Katie Winkle



Abstract
What (or who) does artificial intelligence look like? Does it matter? What are "good" uses of artificial intelligence? Who decides? Robots are slowly moving from novelty to normal, increasingly found on our streets, in our hospitals, schools and homes. At the same time, social inequality is skyrocketing: polarisation is increasing, economic inequality is growing, gender equality is going backwards. Robots have the power to change our behavior, influence our thinking and shape how we interact with each other; will they help solve these problems? Or will they make everything worse? I'll argue that designing robots is really an act of designing society, I believe they really can unlock more equal, joyful and meaningful ways of living, but only if we start thinking (and designing) differently.

Katie Winkle is a docent at the Uppsala Social Robotics Lab at Uppsala University and is a leading researcher in human–robot interaction (HRI). Her interdisciplinary research brings together robotics and AI with psychology, design and the social sciences to explore how interactions with intelligent and autonomous systems can be designed to be trustworthy, socially sustainable and supportive of human flourishing. A central theme in her work is translating principles of trustworthy and responsible AI into the actual design of human–robot interactions. Her research spans participatory design, social influence and behaviour, feminist HRI, robot ethics, interactive machine learning, and the opportunities and challenges that emerge as robots become increasingly embedded in people's everyday lives. Winkle is an active member of the international HRI research community. She has held several prominent roles at the ACM/IEEE International Conference on Human-Robot Interaction, including that her work on Feminist Human-Robot Interaction received the Best Paper Award in Theory and Methods at HRI 2023. At Uppsala University, her research forms part of the university's broader work on trustworthy AI and social robotics, connecting fundamental questions about human cognition, behavior and social interaction with some of the most pressing questions raised by increasingly autonomous and socially capable AI systems.

https://www.uu.se/en/contact-and-organisation/staff?query=N22-79

SweCog 2026 Programme

Monday, 26 October, room 101146, floor 1, house 10, Ångström, House 10, Uppsala University

Time	Activity	Title/Presenter
12-13	Registration opens outside Sonja Lyttkens (room 101121)	
13:15-15 **Workshop: From Brain Signals to Cognitive Claims by Roja Tahamtan, Neuroscientist Lead, InnoBrain**		

Tuesday, 27 October Sonja Lyttkens (room 101121), floor 1, house 10, Ångström, House 10, Uppsala University

Time	Activity	Title/Presenter
08:00	Registration opens	
09:00-09:05	Welcome	Jessica Lindblom
09:05-10:00 **Keynote: Returning to the Elusive Concept of Representative Design by *Peter Juslin*** **Session chair: Jessica Lindblom**		
Session 1: AI, technology and human thinking **Session chair: Jessica Lindblom**		
10:00-10:20	Before the Question: Prompt Generators and the Pre-Structuring of Formulation Work by *Mariam Marlen Mirström*	
10:20-10:40	Help! The mindless AI zombies are running the establishment! By *Joel Parthemore*	
10:40-11:00	Assessing the long term cognitive effects of AI use using GPS navigation as an example by *Nils Dahlbäck*	
11:00-11:15 **Coffee Break** outside Sonja Lyttkens		
Session 2: Language production and speech perception **Session chair: Andreas Bergqvist**		
11:15-11:35	Why ducks “quack!” (and what it says about speech) by *Sonja Dahlgren et al.*	

Time	Activity	Title/Presenter
11:35-11:55	Grammatical encoding of transitive events in Russian is functionally motivated by *Yulia Kashevarova and Thomas Hörberg*	
11:55-12:15	Incremental formant normalization: uniform scaling best explains human vowel perception by *Anna Persson*	
12:15-13:15 **Lunch & Poster Session** at the entrance area of Ångström Laboratory, House 10.		
13:15-14:15 **Keynote: What Would Evolution Build? An Evolutionary Perspective on Associative Memory by *Mats Lind*** **Session chair: Jonathan Källbäcker**		
Session 3: Pragmatics, learning and linguistic context **Session chair: Jonathan Källbäcker**		
14:15-14:35	Turn initiations are subserved by higher-order processes outside of the perisylvian language network by *Caroline Arvidsson*	
14:35-14:55	Adolescent pragmatic development mirrors pragmatic individual differences in adulthood: An fMRI study by *Christoffer Forbes Schieche & Julia Uddén*	
14:55-15:15	Number-line training shapes short-term developmental pathways between approximate and symbolic numerical systems in preschoolers by *Maybí Morell Ruiz et al.*	
15:15-15:30 **Coffee Break** outside Sonja Lyttkens		
Session 4: Interaction, participation and space **Session chair: Mikael Laaksoharju**		
15:30-15:50	Swedish driving examiners' communicative methods during driving tests: Early findings from an ongoing field study by *Daniel Björklund-Flärd and Birgitta Thorslund*	
15:50-16:10	Interpreting 'Here' And 'There' in shared Human-Robot Spaces by *Adwitiya Mandal et al.*	
16:10-16:30	Relational Epistemology of Biological Agency and Artifactual Scaffold by *Gordana Dodig-Crnkovic*	

Time	Activity	Title/Presenter
16:30-16:50	No credible evidence that education protects cognition against brain aging: Converging evidence from causal simulations and longitudinal neuroimaging studies by *Amos Pagin*	
16:50-17:00	End of Day 1 and information about the conference dinner	
19:00 – 23:30	**Conference Dinner** at **Katalin And All That Jazz**	

Wednesday, 28 October 2026 Ångström, House 10, Uppsala University

Time	Activity	Title/Presenter
09:00-10:00 **Keynote: Cognition and emotion in animals: Implications for animal welfare** **by *Linda Keeling and Else Verbeek. Presented by Linda Keeling*** **Session chair: Cora Ljunggren Axelsson**		
Session 5: Olfaction, semantics and aging **Session chair: Martin Hansson**		
10:00-10:20	A data-driven and AI-based approach for mapping the semantic organization of smell by *Thomas Hörberg et al.*	
10:20-10:40	Free odor identification as a tool to study the aging mind by *Jonas Olofsson et al.*	
10:40-11:00	Sex Differences and Female Hormonal Factors in Olfactory Decline and Dementia: A 12-Year Longitudinal Study by *Leila Nathalie Rezazadeh*	
11:00-11:15 **Coffee Break** outside Sonja Lyttkens		
Session 6: Memory, perception and prediction **Session chair: Hannah Gross**		
11:15-11:35	Prediction errors shape episodic memories: Empirical evidence and a novel framework by *Marius Boeltzig*	
11:35-11:55	Reassessing Unconscious Visual Working Memory: Evidence from Contralateral Delay Activity by *Stefan Wiens, et al.*	
11:55-12:15	Patterns of sensory misattribution in spatial memory: A preregistered virtual reality study by *Malina Szychowska*	

Time	Activity	Title/Presenter
12:15-13:15 **Lunch & Poster Session** at the entrance area of Ångström House 10.		
13:15-14:15 **Keynote: Designing Robots, Designing Society by *Katie Winkle*** **Session chair: Hannah Gross**		
Session 7: Context, Interpretation and the Social Mind **Session chair: Andreas Bergqvistt**		
14:15-14:35	A different language, a different self? How second language use shapes risk aversion and personality traits by *Oleksandra Osypenko and Alexandra S. Dylman*	
14:35-14:55	Gaze Allocation During Dynamic Emotion Recognition - An Eye-Tracking Study of Single and Blended Facial Expressions by *Alireza Ranjbar Khanhosseini and Petri Laukka*	
14:55-15:15	The left temporo-parietal junction supports context-based ambiguity resolution by *David Pagmar*	
15:15-15:25 End of SweCog2026 and Closing remarks		

Session 1: AI, technology and human thinking

Before the Question: Prompt Generators and the Pre-Structuring of Formulation Work

by

Mariam Marlen Mirström
Department of Psychology, Lund University, Lund, Sweden

Research on prompting examines how prompt design affects model responses (Gupta et al., 2024) and relates to problem-solving performance and user experience (Choi et al., 2025). Specifying a request is itself work: selecting a role and context, setting constraints, and anticipating a response form (Fuentealba Cid et al., 2024; Reynolds & McDonell, 2021; Velásquez-Henao et al., 2023). Public prompt generators intervene at this stage by turning a minimal input into a more specified prompt. In the present sample, RyRob expanded "Help me improve my academic writing" into a request assigning an academic-editor role, setting editorial standards, and requiring a revised version, change log, and suggestions. These specifications were absent from the input.

In this study, formulation work refers to the activity of articulating a request, including selecting which aspects of the task to make explicit, supplying specifications, and identifying information for later completion. Representational transformation refers to the observable changes in the request's textual specification from the minimal input to the generated prompt, including added specifications and marked fields. This distinction separates formulation as an activity from the textual changes examined in this study.

Surveys and prompt-pattern catalogues organise prompting techniques and reusable patterns (Schulhoff et al., 2024; White et al., 2023). Related work proposes a prompting protocol (Robertson et al., 2024), illustrates applications in entrepreneurial communication (Short & Short, 2023), and examines how non-experts develop and evaluate prompts (Zamfirescu-Pereira et al., 2023). Dalsgaard (2025) conceptualises prompting as cognitive mediation, and Ma et al. (2025) emphasise the continuing importance of human requirement articulation alongside automated prompt refinement. The present study contributes an exploratory analysis of the specifications and marked fields introduced when publicly accessible prompt generators elaborate minimal requests. Each generated prompt is compared with its corresponding input

to identify these additions before the requested answer is produced. Component counts describe the recorded additions and do not assess tool quality.

The analysis uses eight components to describe additions beyond the minimal input. Task specification concerns further detail about the requested operation; role, the stance or expertise invoked; context, relevant situational information; and audience, the intended recipient of the response. Quality criteria concern standards for an adequate answer; constraints, limits on the task or response; process instructions, steps for carrying out the task; and output format, the form of the requested response. A supplied specification provides content for one of these dimensions, while a corresponding marked field identifies information for later completion. Both make a dimension of the request explicit, although the marked field leaves its value open. These components are analytical dimensions, not a checklist of what every good prompt must contain; leaving an aspect open may be appropriate.

Norman (1991) distinguishes the system view, concerning the performance of the person and artefact together, from the personal view, concerning how the artefact changes the task performed by the individual. Together with accounts of distributed cognition (Hutchins, 2001) and the role of external representations in problem solving (Payne, 2001), this distinction motivates examining specifications that users could subsequently inspect, complete, or revise. Norman's distinction provides an interpretive rationale rather than the coding categories. The central claim is that the observed outputs pre-structure formulation work at the representational level by adding specifications and, in some cases, marked fields before the requested answer is produced. This analysis requires no access to internal implementations and provides no evidence about algorithms, training data, or design intentions.

The research question is: which representational components were added to the four minimal inputs in the 16 generated prompts? The analysis concerns particular outputs collected within a specified period; it does not establish stable tool properties or estimate prevalence across prompt generators. User comprehension, learning, agency, authorship, and downstream response quality were not measured. If a generated prompt is adopted, the request reaching the model includes specifications not explicitly stated in the initial input. Identifying these additions provides a basis for subsequent research on authorship and learning in prompt use.

Method

Four publicly accessible text-to-text tools were included: Feedough AI Prompt Generator, Jotform AI Prompt Generator, RyRob AI Prompt Generator, and GeneratePromptAI. Sampling was purposive. Tools were identified through web searches for "AI prompt generator" between

28 April and 17 May 2026 and screened against three inclusion criteria: accepting short natural-language inputs; returning generated text prompts as standalone output; and being accessible without login. Prompt marketplaces, community-curated repositories, and tools embedded within wider AI chat interfaces were excluded. Of six tools screened, four met all three criteria. Taskade required login before generation, while QuillBot displayed generated text within an integrated AI Chat interface. Tool URLs and access dates were recorded. Observations concern versions accessible during this period.

Four minimal inputs were used: “Help me improve my academic writing”; “Help me understand a difficult concept”; “Help me plan a study session”; and “Help me write a professional email”. Each named a broad task while leaving more concrete specifications open. They covered academic writing, concept learning, study planning, and workplace communication, avoiding sensitive domains such as health, politics, religion, legal advice, identity, trauma, and private relationships. All shared the “Help me + verb” syntactic frame.

Each input was submitted once to each tool, yielding 16 generated prompts. The unit of analysis was the first generated prompt for each tool–input pair, compared with its corresponding input. Raw generated texts, including their placeholders, were retained for coding, with outputs associated with their source tool and minimal input. Follow-up questions, regeneration options, and offers of multi-step refinement were noted but excluded from coding, keeping the analysis at the initial artefact and avoiding selection among outputs. No tone, style, audience, or advanced settings were adjusted. Where platform selection was required, ChatGPT was selected if available; otherwise, the default or most general text-generation option was used. No repeated generations were collected. The eight-component scheme combined deductive and inductive categories. Role (corresponding to persona), context, constraint, and output format were deductively informed by prompt-pattern and prompt-engineering taxonomies (Schulhoff et al., 2024; White et al., 2023). Task specification, audience, quality criteria, and process instruction were retained inductively when they recurred in the outputs and captured distinct additions not adequately represented by the initial categories. The scheme was developed and applied within the same corpus; these frequencies do not independently validate it as a taxonomy.

Coding concerned specifications added beyond the minimal input. Codes were 0 = not added; 1 = added but generic or transferable; and 2 = added and task-linked in the observed output. Each generated prompt received one code for each of the eight components. Restating information already in the input did not count as an addition. Code 2 indicates that the added component was semantically tied to the input domain rather than transferable unchanged to an

unrelated task. Task-linked denotes domain specificity, not quality. Task specification required a more concrete action beyond the minimal request; quality criteria required standards for an adequate response. For a worked example, RyRob's output for "Help me improve my academic writing" began: "Act as a professional academic editor with a PhD in [FIELD OF STUDY]." Here, "professional academic editor" exemplifies a task-linked role (code 2), while "[FIELD OF STUDY]" marks a discipline to be specified later. The same output specified a task-linked quality criterion (code 2): "Tone: Use a formal, objective, and scholarly voice." By contrast, "helpful assistant" as a role or "clear answer" as a quality criterion would, on its own, be generic (code 1). The example illustrates how supplied specifications and open fields can coexist within a generated prompt. After the main coding, the number of generated prompts containing at least one explicit placeholder was counted descriptively from the retained raw texts. A placeholder was narrowly defined as a marked slot, such as [INSERT TEXT], [FIELD OF STUDY], or [RECIPIENT ROLE/TITLE]. This post hoc count was separate from the eight-component matrix. In the component coding, when a placeholder introduced a field beyond the input corresponding to one of the eight categories, that component was coded as added; the codes did not distinguish a supplied value from an open slot. Placeholders for the task object, such as [INSERT CONCEPT] or [INSERT EMAIL PURPOSE], were not coded as context. The author coded the outputs, drawing on initial coding suggestions from a language model. Ambiguous cases were reviewed against the decision rules before the codes were finalised. No inter-rater reliability estimate was calculated.

Results

Task specification and quality criteria were coded as task-linked additions in all 16 prompts. Other components were added in 8 to 13 outputs (Table 1). Across the 128 component codes (16 prompts × 8 components), 88 indicated task-linked additions, 8 generic additions, and 32 no addition beyond the input. Of the 32 component codes per tool, 32 indicated additions for Feedough, 30 for RyRob, and 17 each for Jotform and GeneratePromptAI. Equal totals nevertheless reflected different component patterns: role was added in all four GeneratePromptAI outputs and one Jotform output, whereas process instructions were added in two Jotform outputs and no GeneratePromptAI outputs. For the professional-email input, output format was added in all four outputs. Process instructions were added in 9 of the 16 outputs. The post hoc count identified at least one explicit placeholder in 11 of the 16 prompts. Examples included [FIELD OF STUDY], [INSERT TEXT], [INSERT CONCEPT],

[SUBJECT], and [RECIPIENT ROLE/TITLE]. This count concerns outputs containing at least one marked slot, rather than the number of component instances left open.

Table 1. Added formulation components across 16 generated prompts.

Component	Added /16	Task-linked	Generic	Not added
Role	13	13	0	3
Task specification	16	16	0	0
Context	12	10	2	4
Audience	11	8	3	5
Quality criteria	16	16	0	0
Constraint	8	8	0	8
Process instruction	9	7	2	7
Output format	11	10	1	5

Note. Added combines codes 1 and 2. Task-linked = code 2; generic = code 1; not added = code 0. Each prompt received one code per component. Prompts with at least one marked placeholder, out of four per tool: Feedough 4, RyRob 4, Jotform 2, and GeneratePromptAI 1.

Discussion

The analysis identifies which aspects of minimal requests were elaborated at the prompt-generation stage. Every recorded output added task specification and quality criteria, introducing further task details and evaluative standards. Of the 96 added component instances, 88 were coded as task-linked, indicating that most additions related to the input domain. Ma et al. (2025) found that a prompt optimiser could modify requirements, while emphasising the continuing importance of human articulation of customised needs. In the present study, domain-related elaboration does not establish a match with a particular user's intended requirements. The marked fields suggest an analytical distinction between filling and slotting. Filling supplies a specification absent from the input, such as a requirement for a scholarly tone; slotting introduces a field, such as [FIELD OF STUDY], while leaving its value open. A slot thus selects a dimension for further elaboration even when it supplies no value. Under the 0/1/2 scheme, an added component could contain supplied specifications, applicable open fields, or both. The count of 11 prompts with marked slots does not establish how often each component was filled or slotted, and not every placeholder corresponds to one of the eight components. Component counts therefore cannot establish how fully the request was specified. Future coding could add a separate distinction between supplied, open, and mixed specifications within each component, while retaining cases with no addition. Accounts of cognitive artefacts (Norman, 1991) and distributed cognition (Hutchins, 2001) motivate examining how these textual differences may organise subsequent activity. For example, users

could inspect a supplied specification, complete or revise a marked field, or identify a relevant dimension left unelaborated. A follow-up could hold a minimal request constant while varying whether a component is supplied, presented as an open field, or left unelaborated. Observations of what participants notice, accept, and revise, together with their explanations of specifications, could examine comprehension and use. Changes from baseline in the appropriate use of specifications on a new unaided task could provide evidence of learning. Attributions of authorship and ratings of felt agency could examine perceived contribution and control; subsequent model responses could be evaluated separately. The study is limited by its purposive four-tool sample, four similarly phrased inputs, single generation per tool–input pair, and single-author coding informed by language-model suggestions, without an inter-rater reliability estimate. Repeated generations may differ, and public tools may change following updates. Generalisation to fragments, declarative requests, or role-prefixed requests remains untested. The non-sensitive domains restrict the formulation demands examined, and the inputs do not separate topic from genre. The contribution is an exploratory account of how generated prompts pre-structure requests through supplied specifications and marked fields, providing distinctions for subsequent research on formulation in use.

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Included tools: feedough.com/ai-prompt-generator; jotform.com/ai/prompt-generator; ryrob.com/ai-prompt-generator; generateprompt.ai/en. Access dates fell within the period reported above.

Help! Mindless AI zombies are running the establishment!

by

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"Intelligence", for purposes of this paper, is the ability – associated in the first instance with living organisms – to interact with the world in a flexible manner that, in the extreme, is indicative of consciousness and conceptual agency (which, I have argued previously (Parthemore, 2017, p.37), are two sides of one coin). It is cognition that builds on learning and problem solving. This paper argues that humanly intelligent artefactual agents are not, despite claims to the contrary, on the horizon – never mind, as some would have it, already with us. Neither are existing AI agents conscious nor, in any useful sense of the word, cognitive agents. The hype around generative AI obscures the real dangers:

- The way generative AI systems are designed to come across as smoothly polished and self-confident in a way that mimics the behaviour of aggressively dominant individuals who tend to get their way (Koravamode, 2023).
- The way these systems are known to reflect biases inherent in the training corpora and which they essentially summarise over in giving their responses (Afreen *et al.,* 2025).
- The increasing dependence of people – even highly trained researchers in cognitive science – on tools known to be biased and otherwise flawed in their output (Kooli *et al.,* 2025).
- The tireless capacity of generative AI systems to produce misinformation and misleading information that floods social media (Zhou *et al.,* 2023).
- The increasing reliance of many, partly in response to the growing media deserts and partly due to scepticism toward traditional media, to rely on social media and other non-traditional sources as their primary news sources (Knutson *et al.,* 2024).
- The way the technology encourages people to stop thinking critically, allowing them to be guided by pseudo-agents that merely seem wise and benevolent (Lee *et al.,* 2025).
- The lack of adequate guardrails on the technology and the fierce resistance to such guardrails from the industry (Rauh *et al.,* 2024).
- The way that, in pursuit of profit, human workers are replaced by AI models (see, e.g., Le & Huynh, 2026) that cannot work to the same standard of performance (witness automated filtering of CVs and replacement of call centres by AI agents).

Many have claimed that generative AI is either near to, or already crossing over into, conscious intelligence – capable of reflection, capable of authorship, perhaps capable of feelings (see Seth, 2026). If that claim is, as I argue, a mistake, what gives rise to it; and what are the consequences?

AI has, since its inception in the 1950s, pursued two distinct paths: one focusing on AI as ***artificial*** intelligence with a focus on reproducing behaviours that previously were thought to require (human) intelligence but turn out not to. This is AI as engineering and automation, as described by Blay Whitby (2003). The archtype, perhaps, is Joseph Weizenbaum's (1966) Eliza which, to Weizebaum's shock and dismay, convinced many that they were engaging in a meaningful conversation with a thoughtful, intelligent partner – even to point of being upset if the program was suddenly terminated.

The other path has pursued artificial ***intelligence***: which is to say, the (re-)creation of mind ("artificial general intelligence" or AGI), often coupled to the idea that mind is computational

with rarely, if ever, any attempt to define what "computational" means except to commit to a mechanistic view of mind. Typically, there is a commitment to one of the two following positions:

- Intelligence is all about a ***lot*** of explicit rules, together with symbol manipulation (the originally dominant approach), or…
- Intelligence is all about a ***few*** explicit rules and a great deal of pattern matching based on a training set (the currently trending connectionist or machine-learning approach).

Generative AI represents the first kind of AI research masquerading as the second. The reasoning appears to be, make the presentation sufficiently convincing, and for all practical purposes one ***is*** conversing with a thoughtful, intelligent partner. The problem is that, with knowledge of what to look for, the illusion quickly can be made to break down.

Generative AI models are poor at solving problems that require reasoning about one's own reasoning: so-called *meta-reasoning* (Yoshizawa *et al.,* 2026). For much the same reason, they cannot distinguish real from hallucinated: that requires reflection. They are poor at solving novel problems, even if people find them relatively easy to understand. After all, novel problems are reflected nowhere in the training corpora. They are likewise poor at solving novel formulations of familiar problems, if not sufficiently close to the original formulation to be "recognisable" through statistical analysis. Both tasks require analogical reasoning: the capacity to draw parallels to past experience (Qin *et al.,* 2025). Most tellingly, perhaps, they struggle to develop an argument or structure a longer text in a way that makes more than superficial sense (Malhotra, 2025). It sounds good, but what (on reflection) are they really saying? The presentation will offer concrete examples of all these deficiencies.

It is telling that, as much as these systems are widely described as intelligent in a human-like way or even conscious, few if any are ready to commit to calling them alive. Given that all our experience of human-like (as opposed to artificial) intelligence and all our experience of consciousness is derived from living creatures, it behooves the proponents of these claims to give some plausible explanation why intelligence – or consciousness – should, nevertheless, be attributed. Mere mimicry should not be sufficient: a model and that which is modeled are two quite different things. When does a model stop being a model?... perhaps, like Collodi's puppet, when it comes to life.

In setting out his "semiotic hierarchy", Jordan Zlatev (2018, 2009) offers a non-biochauvinist way of justifying the intuition that ("true") intelligence requires life (even if that life is, like the original robots from Čapek's play, not naturally evolved but grown in a lab). The strictly

functional definition of life offered by *autopoiesis* (Varela *et al.,* 1974) shows how this might work. Zlatev describes true language fluency – as opposed to mere linguistic competence – as an embodied experiential activity dependent on semiosis, itself in turn dependent on consciousness – which is dependent on cognition, which is dependent on life. From language down to life, each level presupposes the next. For Zlatev, too many otherwise competent researchers are buying the hype and failing to look more closely.

If the arguments presented here are correct, the ethical and practical implications are significant. Generative AI models are the palest reflection of their creators, *AI as engineering* pretending to be *AI as AGI*. They are imitators *par excellence,* utterly mindless but none the less destructive for it. Students are using them to write papers without the means to tell whether those papers are meaningful or gibberish. Academics are under intense pressure to use them: to write their lesson plans, grade student submissions, even help them write their own papers. We are dumbing ourselves down, re-creating ourselves in the image of our own creation, even as our species faces an existential threat from climate change fueled in part by the AI agents' insatiable hunger for energy.

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Assessing the long-term cognitive effects of AI use using GPS navigation as an example

by

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The use of AI has exploded in recent years, as has the discussion of the long-term effects of use of AI. But also insightful discussion is hampered by lack of empirical evidence (e.g. Ke *et a*l.2026 and Pritchard *et al* 2026) for the obvious reason the devices like ChatGPT has not been around long enough to provide an empirical base for its long-term effects.

But GPS-based navigation devices have been available for public use since 2000 and are since the advent of smart phones about a decade later freely available and frequently used by most lay people to navigate to destinations when driving and walking. This gives an opportunity to draw potential explanations rather hypotheses that might also apply to today's freely available AI devices and perhaps suggest ways forward for the use of AI in areas like higher education.

Long-term effects of GPS use on spatial cognitive and wayfinding abilities are reviewed in a section of Mingon, Gillet, and Sutton (2026), and the results are clear. Two examples: Minaei (2014) found in a large-scale ethnographic study that people that used GPS had worse map-knowledge of London than people who did not use GPS. Ishikawa (2019) studied people with varying experience with and use of GPS, from never to 20 years, and both car and walking, and found a negative correlation between sense of direction GPS use. This pattern was reversed for paper map-users – those that used maps the most had the best sense of direction.

The development of a reasonable accurate mental map (survey knowledge) cannot be achieved without experiences with using actual maps. But when the maps are digital and combined with a GPS the positive effects of external representations on our internal cognitive abilities do not hold any longer.

The "cognitive decline" of using GPS seems at first appearances to go against a fundamental result in cognitive development research, namely that the development of our internal cognition is changed and often enhanced by the use of external representational systems (e.g. Norman, 1993, Vygotsky's 1987, Luria 1976). How can this be explained, and can this explanation also hold for the use of other "intelligent" devices?

One explanation, inspired by Craik and Lockhart (1972) and Ericsson et al (1993), is that it is necessary to be cognitively active in the use of an external representation for it to have positive effects on internal cognition. With a paper map you are, with a GPS device you are not. They

do not enhance or support cognitive processes; they take over them. And this seems to hold also for most so-called AI devices.

This in turn suggests that the potential negative effect of AI use in e.g. education is not primarily that students not learn facts, but that they do not learn how to process facts. It is the cognitive processes that are not trained and developed in our students. Which in turn suggests that higher education needs to shift its focus from factual knowledge to the development of the cognitive skills.

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Session 2: Language production and speech perception

Why ducks “quack!” (and what it says about speech perception)

by

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Introduction

In linguistics, iconicity refers to motivated relationships between linguistic form and meaning. This short paper asks why words for the duck calls converge across languages, and whether that convergence tracks measurable properties of the call itself. Traditional paradigms such as the classic “kiki-bouba effect” demonstrate that people reliably associate sharp-sounding pseudowords with angular shapes and round-sounding pseudowords with soft shapes (Köhler, 1929; Ramachandran and Hubbard, 2001). Modern work on the topic has expanded the frame of reference for linguistic iconicity, giving rise to a vibrant new community of cognitive linguistics. Here, we argue that onomatopoeia – especially animal-noise words – offer a uniquely tractable natural laboratory for studying how real-world acoustic signals in the environment become stable lexical items (Ekström & Dahlgren, 2026). Specifically, because the external referent – i.e., the animal call – is fixed, language independent, and perceived comparably by listeners across cultures, animal onomatopoeia allows direct comparison of perceptual, imitative, and representational processes across languages and cultures.

Despite this, even distinctly onomatopoeic words for animal calls show substantial cross-linguistic variation – likely shaped by phonological inventories, and language-specific imitation biases (Dahlgren & Kittilä, in prep.). Dog barks, for example, appear predominantly as “Hau-type” diphthongal forms in languages with large vowel inventories but shift toward velar or approximant types in languages with, in comparison, more extensive consonant inventories; pig grunts likewise display multiple guttural and nasal-velar patterns with clear

areal clustering. Here, we present a case study of vocalizations by the common wild duck or mallard (*Anas platyrhynchos*), and highlight how such work can bear directly on broader questions in speech perception, language evolution, and iconicity research: how listeners extract stable acoustic qualities from environmental sounds, how speakers' phonological systems filter those imitations, and how the resulting forms stabilize as conventional words in lexical databases. By focusing on this low-variation exemplar, the paper highlights a key mechanism in iconicity: certain environmental sounds appear "pre-adapted" for lexicalization because their acoustic properties align closely with universal biases in human speech production and perception (Ekström & Dahlgren, 2026). This alignment minimizes the role of language-specific phonological constraints that dominate in more variable calls such as dog barks or pig grunts. The research question is therefore: do the acoustic features of wild mallard quacks align with the speech sounds that dominate duck words across languages?

Methods

Lexicographic data

The study is a comparative corpus analysis plus a qualitative and quantitative acoustic description. There were no human participants, no listener experiment, and no laboratory site. Onomatopoeic forms were drawn from the cross-linguistic database of animal-noise words compiled by Dahlgren and Kittilä (in prep.). The duck subset comprises forms attested in dictionaries and native-speaker sources for 21 languages, not elicited from listeners in this study. Each form was coded into one of two families: Kwak-type (velar or labial-velar onset, open vowel, velar coda) or plosive-type (voiced plosive onset, open vowel, optional plosive coda). In this sample, duck calls exhibit strikingly marginal variation. Forms cluster into two dominant families: First, a "Kwak type", characterized by an initial velar stop [k] and approximant [w] followed by a low or mid-low vowel and a concluding coda velar. Examples include the English *quack quack* [kwæk], Finnish *kvaak-kvaak* [kvaːk], French *coin coin* [kwɑ̃], Swedish *kvack-kvack* [kvɑːk]). Second, a smaller "Plosive type", characterized by a plosive [g], followed by a low vowel. Examples of this form include the Japanese *ga ga* [ga]. The form may also include a final plosive, such as the Turkish *gak gak* [gak]. Across languages, then, patterns of vowels and plosive/velar sounds are highly consistent. This uniformity is remarkable given the phylogenetic and geographic spread of the sampled languages (spanning Indo-European, Uralic, Dravidian, and others) and contrasts sharply with the higher variability observed for dogs and pigs in the same dataset (Dahlgren & Kittilä, in prep.). The present short paper examines why duck quacks converge on such similar forms.

Acoustic data

A representative sample of 100 wild mallard quacks was downloaded from the bioacoustics archive *xeno-canto* (https://xeno-canto.org/). Recordings were drawn from the "A" (highest-quality) category and were not noted as distant from the microphone. Spectrograms were generated in *Audacity* (audacityteam.org). Because no prior phonetic protocol for duck calls exists, the first (F_1) and second formants (F_2) (corresponding to acoustic energy peaks, and a key feature of vowel production) were estimated by one author (AE).

Results

Across the 21 languages, duck words fall into two families only: a Kwak-type [k/kw + open vowel + velar coda], as in English quack, Finnish kvaak, French coin, and Swedish kvack, and a plosive-type [g + open vowel, with or without a coda], as in Japanese ga ga and Turkish gak gak. This is contrasted by the higher variation for dogs and pigs in the same database.

Formant estimates for duck *quacks* were F_1 = 1191 Hz, and F_2 = 1853 Hz. In comparison, Catford (2001) reports averages of F_1 = 850 Hz and F_2 = 1610 Hz for male speakers for open unrounded vowel [a]. For consonants, in natural speech, velar stop /k/ shows more rapid and concentrated energy in spectrograms because its oral closure allows high supraglottal pressure to build and release explosively as a sharp, broadband acoustic burst, whereas the glottal stop /ʔ/ lacks this oral pressure build-up and instead releases via gradual vocal-fold opening. As a point of rough comparison, the sudden "burst" of energy exhibited by duck calls is more consistent with the first. As such the concentration of sonorant acoustic energy in duck calls markedly overlaps with that observed in the speech sounds typically used to imitate them in natural languages. In particular, the relatively low-frequency energy concentrations and rapid spectral transitions align with known perceptual sensitivities that facilitate the mapping of environmental sounds onto speech-like forms. This alignment explains why the corresponding onomatopoeic words are perceived as natural matches for the actual quack.

Discussion

The cross-linguistic data reveal remarkably low variation in duck onomatopoeia compared with the high diversity observed for dog barks and pig grunts in the same dataset (Dahlgren & Kittilä, in prep.). Across 21 languages spanning Indo-European, Uralic, Dravidian, and others, forms converge on either a Kwak-type pattern (English quack, Finnish kvaak, French coin,

Swedish kvack) or a plosive-type pattern (Japanese ga ga, Turkish gak gak), and not on a third family. This uniformity is not predicted by areal or genetic factors alone and stands in sharp contrast to the areal effects and phonological filtering evident in other animal calls (for example, Hau-type diphthongal forms dominant in dog barks in vowel-rich languages, or guttural forms for pigs).

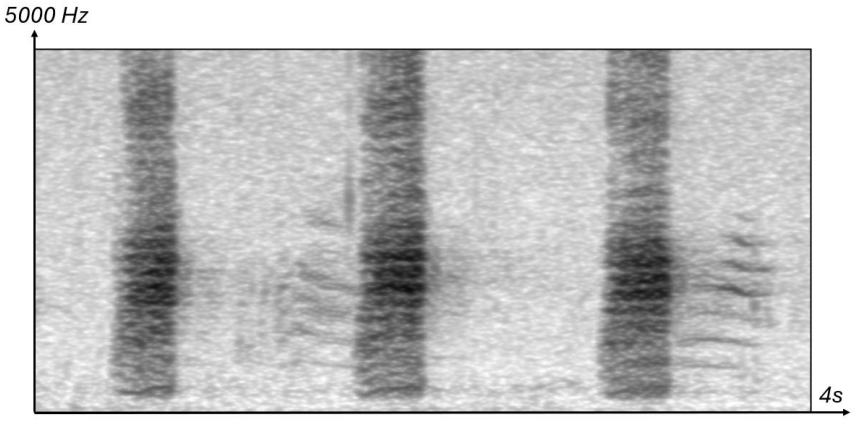


Figure 1. Spectrogram of three common "quack" vocalizations by a wild mallard (*Anas platyrhynchos*). The sudden onset is lexicographically imitated almost exclusively as a "kwak" [kVk] or [kwVk] or "plosive type" such as [gV], [gVk] or [mVk]. In both cases, V corresponds to an open or mostly open vowel.

We argue that this convergence arises because the acoustic signature of the wild duck call (prominent low-frequency energy, rapid spectral transitions, and noisy amplitude bursts) maps especially well onto the human vocal tract's natural production and perceptual repertoire. Low vowels provide an optimal match for the call's F_1-F_2 structure, while velar/approximant onsets and plosive releases capture the abrupt onsets and broadband noise. These phonetic features are not only easy to produce but also highly salient to listeners, thereby facilitating rapid imitation and conventionalization "from the sound up."

This case therefore illuminates a potential key mechanism in iconicity research: certain environmental sounds are more amenable to lexicalization because their acoustic properties align closely with universal biases in speech perception and production (cf. Imai & Kita, 2014). Unlike more variable calls such as dog barks or pig grunts, the duck quack appears to minimize the role of language-specific phonological inventory constraints. Because cross-linguistic variation is already minimal, language-specific inventory effects are smaller than for barks or grunts, so a match between call and word is easier to attribute to shared production and perception biases rather than to areal borrowing or the shape of a particular phoneme inventory. That is what makes the case clean: not that the sound is intrinsically special, but that competing linguistic explanations are weaker here than for more variable calls. For a cognitive-science

audience, the point is how listeners extract stable features from an environmental sound, and map those features onto speech gestures already in the human repertoire, and how iconicity can then scaffold a conventional word.

Several limitations should be noted. Experimental validation is needed to confirm that listeners indeed perceive onomatopoeic forms as better matches for real duck calls than arbitrary labels. Future studies could usefully extend the approach to behavioral matching tasks or manipulated stimuli to isolate the precise acoustic drivers of the observed uniformity. In sum, the marginal variation in duck quacks provides compelling evidence that speech-perception and production biases actively shape iconic word forms.

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Grammatical encoding of transitive events in Russian is functionally motivated

by

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Abstract

Transitive sentences describe events with two participants – the one most responsible for the event occurrence and the one affected by it. Various information helps comprehenders assign grammatical functions that encode participants of an event (as subjects and objects) and typically correlate with their roles in the event. The information use in written Swedish has been found to be functionally motivated: writers use a potentially ambiguous object-first word order less frequently if disambiguating information, such as animacy, is unavailable. They also provide pragmatic information that reflects noun phrase (NP) referents' accessibility in the discourse model (e.g., givenness) so that given and thus highly accessible information is positioned sentence initially. We investigate whether similar patterns hold for Russian subject-verb-object (SVO) and object-verb-subject (OVS) sentences. Using corpus data, we analyse the distribution of animacy and givenness of subject and object NPs. We find evidence in support of the functional account also in Russian.

Background

Evidence suggests that natural language communication is shaped by speakers' desire to balance communicative success and production ease during on-line sentence production (e.g., Hörberg & Sjons, 2023). Accordingly, a linguistic expression should be informative enough for a recipient while being as economic to produce as possible. From this perspective, the grammatical encoding choices language producers make are functionally motivated. This hypothesis may be investigated by analysing the distribution of information in natural language use.

Hörberg (2018) found evidence for these functional motivations underlying the encoding of grammatical functions (i.e., subject and object) in written Swedish transitive sentences. While the dominant word order in Swedish transitive sentences is SVO, an OVS order is also possible; and since Swedish only case-marks pronominal NPs (e.g., *Flickan/henne gillar jag* 'The girl/her-OBJ I like'), OVS sentences with lexical objects may be ambiguous with respect to grammatical functions. In such sentences, the animacy of the sentence-initial

NP object serves as a cue to grammatical functions, facilitating interpretation (Hörberg & Jaeger, 2021). Using corpus data, Hörberg (2018) found objects in potentially ambiguous OVS sentences to be more frequently inanimate, indicating that writers balance their use of morphosyntactic and semantic information in order to be informative enough for the reader, in terms of avoiding the OVS word order when the object NP referent is animate.

The word order choice in a transitive sentence is often guided by discourse pragmatic considerations. In head-initial languages like Swedish and Russian, there is a preference to place entities that are accessible in the discourse model (e.g., higher on the givenness scale, easily retrievable from the immediate context, see Ariel, 1990) sentence-initially. Indeed, Hörberg (2018) found that objects in Swedish OVS sentences are given more frequently than post-verbal objects. He interpreted this finding to indicate that the OVS word order is used to signal that the object is the discourse topic, i.e. what is currently being talked about in the discourse and thus is highly accessible.

Here, we ask if similar findings as those of Hörberg (2018) also hold for Russian, a language with more elaborate case marking and more flexible word order. Investigating the distribution of object NP properties in Russian SOV and SVO sentences, Slioussar and Makarchuk (2022) found that pre-verbal objects were more often given and pronominal than post-verbal objects. However, to our knowledge, no study has investigated the distribution of both subject and object NP properties, in Russian SVO and OVS sentences, and, to what extent those distributional patterns are functionally motivated in terms of ambiguity avoidance and discourse pragmatic considerations. While Russian unambiguously case-marks pronominal and lexical NPs, marking of some inanimate and a few animate lexical NPs can be uninformative due to the forms' syncretism (i.e., identical nominative and accusative forms). In 1a and 1b below, the word order is OVS, and the grammatical functions of the initial NPs can therefore not be determined on the basis of word order. In 1a, the initial NP is inanimate, allowing for an OVS interpretation at the onset. In 1b, on the other hand, the initial NP is animate and therefore semantically ambiguous with respect to its grammatical function. However, in contrast to Swedish, the grammatical function can still be determined on the basis of case marking.

1a) *Стойку осаждала очередь.*
counter-SG.ACC besiege-PST.SG queue-SG.NOM
'The counter was besieged by the queue.'

1b) *астронавтов будет ждать лаборатория*
astronaut-PL.ACC be-FUT.3SG wait-INF laboratory-SG.NOM

‘A laboratory will be waiting for the astronauts.’

In contrast to Swedish, ambiguity in Russian OVS sentences with animate objects is thus rare. We therefore expect objects of OVS sentences to be animate equally frequently as in SVO sentences.

If the Russian OVS word order is used to signal that the NP object is discourse topical as in Swedish, we also expect NP objects to be more frequently given in OVS than SVO sentences.

Method

A part of the balanced written treebank corpus ‘SynTagRus’ (Droganova et al., 2018) was used. The corpus has over a million tokens in a range of popular science, contemporary fiction texts, journal and newspaper articles, dated 1960–2020. Transitive clauses (n=1418) with nominative subjects and accusative or genitive objects were extracted from a part of the original CoNLL-U files using Python and manual checking. Subject and object NPs were manually annotated for givenness and animacy. Givenness was derived from NP forms (e.g., Gundel et al., 1993) and had two degrees on the scale: less accessible (‘new’) entities could be expressed by lexical NPs and accessible (‘given’) NPs – by pronouns. Following Hörberg (2018) and Dahl and Fraurud (1996), human, non-human animate entities and organisations which act volitionally were annotated as animate.

Results

For animacy, we investigated the distribution of NP animacy across word orders (SVO vs. OVS) and grammatical functions (subject vs. object). We excluded pronominal object NPs as these are unambiguous independent of animacy. The results are shown in Panels A and B. We ran logistic regression models in *R* (R Core Team, 2025) predicting NP animacy as a function of word order, grammatical function and their interaction. Whereas subject NPs were found to be more often animate in SVO (78%) than in OVS (60%) sentences, $\beta = -0.87$, $Z = -4.32$, $p < .0001$, object NPs were animate similarly frequently in SVO (10%) and OVS (13%) sentences, $\beta = 0.32$, $Z = 0.86$, $p = 0.388$. The difference was also highlighted by the significant interaction effect, $\beta = 1.19$, $Z = 2.83$, $p < .05$.

For givenness, we again investigated the distribution of NP givenness across word orders and grammatical functions, this time including both lexical and pronominal object NPs. These results are shown in Panels A and C. We ran logistic regression models, predicting NP givenness from word order, grammatical function and their interaction. As highlighted by their significant interaction, $\beta = -3.6$, $Z = -10.96$, $p < .0001$, there is a reciprocal relationship between word order and grammatical function with respect to givenness. Both subjects (52% vs. 17%,

β = 1.69, Z = 6.58, p < .0001) and objects (50% vs. 13%, β = 1.91, Z = 9.34, p < .0001) are much more frequently given in the sentence initial position.

Discussion and conclusions

The results confirm our hypotheses and suggest that functional motivations similar to those in Swedish guide the morphosyntactic encoding choices Russian language writers make in their production of transitive sentences. In contrast to Swedish, the choice between SVO and OVS word order in Russian seems not to be influenced by the animacy of the (lexical) object NP. In Swedish, OVS sentences with animate object NPs are avoided, presumably because language producers are inclined to avoid locally ambiguous sentences (Hörberg, 2018; Hörberg & Jaeger 2021; Hörberg & Sjons 2023). However, in Russian, animate lexical NPs are predominately unambiguously case marked, and OVS sentences with initial animate objects are thus rarely ambiguous. Our findings are therefore in line with the predictions of a functional account. We also found NP arguments to be more frequently given when positioned sentence-initially, independent of their grammatical function. This is in line with the assumption that NP referents that are highly accessible in the discourse model, and thus high in givenness (Gundel et al. 1993), tend to be positioned sentence-initially. In addition, these findings suggest that the OVS word order in Russian is used to signal that the object NP is accessible and given (i.e., topical) in the discourse model, similar to what has been found in Swedish (Hörberg, 2018).

Ongoing work and contributions

This study is a part of an ongoing project investigating pragmatic and functional motivations behind the writer's choice of each of the possible six word orders and less typical transitive clauses in Russian, encompassing a large set of semantic, pragmatic and morphosyntactic information types. It also includes two studies investigating the processing of transitive sentences of multilingual speakers.

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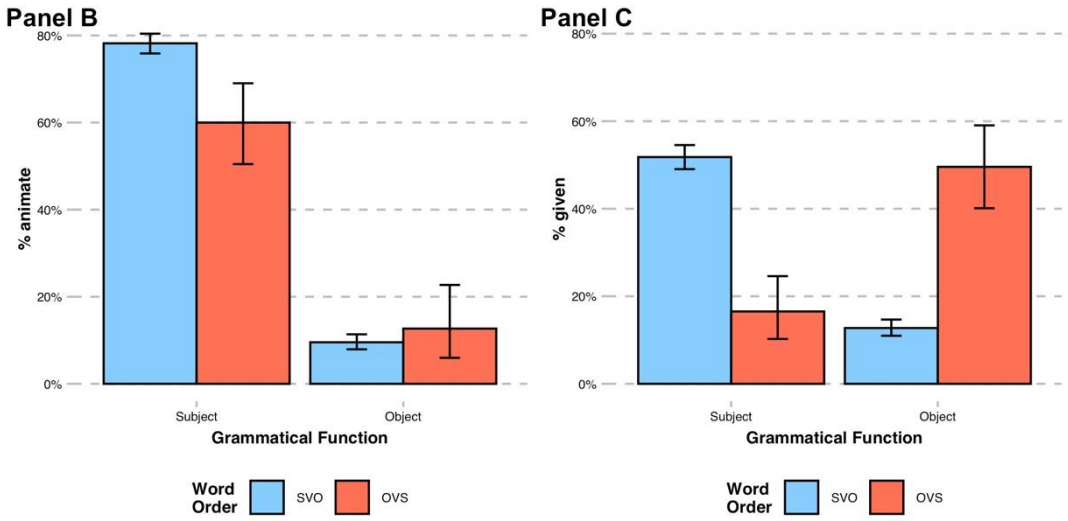
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Panel A. The distribution of animate and inanimate subject NPs and non-pronominal object NPs and given and new subject and object NPs in SVO and OVS sentences.

		SVO		OVS				SVO		OVS	
		n	%	n	%			n	%	n	%
	Animate	1019	78%	69	60%		Given	675	52%	19	17%
Subject	Inanimate	284	22%	46	40%	**Subject**	New	628	48%	96	83%
	TOTAL	1303	100%	115	100%		TOTAL	1303	100%	115	100%
	Animate	112	10%	9	13%		Given	166	13%	57	50%
Object	Inanimate	1062	90%	62	87%	**Object**	New	1137	87%	58	50%
	TOTAL	1174	100%	71	100%		TOTAL	1303	100%	115	100%



Percentage of animate **(Panel B)** and given (**Panel C**) NPs by grammatical function and word order.

Incremental formant normalization: uniform scaling best explains human vowel perception

by

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Listeners recognize vowels robustly despite substantial cross-talker variability in formant realization. This robustness is attributed to auditory normalization mechanisms that correct for individual differences in vocal tract physiology. Over 100 normalization accounts now exist, differing in computational complexity and the amount of acoustic information assumed to be tracked by listeners. The most parsimonious accounts include static transformations of formants into a psychoacoustic space (e.g., Mel, ERB, Bark). The most complex accounts rely on formant statistics tracked and aggregated across all speech tokens from the same talker (e.g., Gerstman, 1968; Lobanov, 1971).

Crucially, past comparisons of accounts have relied on the unrealistic assumption that listeners have immediate access to all relevant talker-specific properties, even from the first observed speech input. In reality, however, listeners must infer these properties incrementally from talkers' speech.

Contribution. We developed incremental variants of eleven influential normalization models—differing in assumptions made about listeners' inference and memory capacities. We then compared these eleven models against the standard, non-incremental models of the same type, and against each other, in terms of their ability to explain human perception.

Study 1. We evaluated cross-validated predictions of each model against US English listeners' responses in two 8-way forced-choice vowel categorization experiments using either natural or synthesized *hVd* stimuli (data from Persson et al., 2025). In contrast to previous work, both experiments used speech from a single talker, allowing listeners (and models) to benefit from information accumulated across trials. We present results based on F1 and F2.

Overall, incremental variants outperformed the standard non-incremental variants, confirming that listeners need to *infer* talker properties from incoming data. Across both natural and synthesized speech, the best-performing incremental models were variants of computationally simple, *single-parameter*, uniform scaling accounts (Johnson, 2020; Nearey, 1978). These accounts outperformed more complex Lobanov normalization, which remains the standard in many fields of linguistics.

Study 2. We compared the winning models from Study 1 against a Bayesian implementation of uniform scaling (the *probabilistic sliding template model* or *PSTM*, Nearey & Assman,

2007). The PSTM is a one-shot normalization model that jointly infers the vowel category *and* talker physiology *from a single vowel instance*, by taking into account the covariance between talkers' f0 and formants (Barreda & Jaeger, 2025). The PSTM outperformed other normalization accounts in previous evaluations against multi-talker experiments. In contrast, we found that the PSTM underperformed on our single-talker experiments.

Conclusions

Our results lend further support to the hypothesis that formant normalization relies on simple, uniform scaling rather than more complex computations (Study 1; also, Barreda, 2020, 2021; Persson et al. 2025), that this scaling changes incrementally (Study 1), and accumulates across observed vowel instances (Study 2). As both studies relied on single-talker input, further work is needed to establish whether these conclusions generalize to more variable, multi-talker listening conditions. Future research should also develop and test cumulative versions of the PSTM to determine whether the one-shot limitation of existing PSTM implementations is indeed the reason for its inability to explain listeners' perception in our single-talker experiments.

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Session 3: Pragmatics, learning and linguistic context

Turn initiations are subserved by higher-order processes outside of the perisylvian language network

by

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Conversation is arguably one of the most ubiquitous contexts for producing language. It is characterized by turn-taking, in which speakers rapidly alternate between speaker and listener roles, while simultaneously planning their upcoming turn. Neurobiological models of speech preparation have implicated regions within the left hemispheric-dominant classical perisylvian language network (e.g., the inferior frontal gyrus, the superior temporal lobes, and the parietal cortex), as well as lower-level sensory and motor areas. These models, however, are grounded in the production of single linguistic units (e.g., words) in non-interactive contexts. Consequently, the neural substrates subserving the initiation of conversational turns are largely unknown. In this fMRI investigation, we asked whether turn initiations would activate areas outside the classical perisylvian core language network. We hypothesized that the initiation of conversational turns would engage cortical regions associated with higher-order processing outside of the perisylvian language network. We analyzed two different conversational datasets, in which participants engaged in unscripted conversations via an audio-video link with a confederate outside the scanner: (1) the Neuroengage dataset (28 participants, comprising 14 hours of conversation; Torubarova et al., 2024) and (2) the Multimodal Corpus of Bidirectional Conversation during fMRI Scanning (24 participants, 4.8 hours of conversation; Rauchbauer et al., 2020). Turn initiations were defined as a 600 ms time window (based on the reported minimum latency of speech preparation; Indefrey & Levelt, 2004), preceding the participant's turn. For each data set, a general linear model was used with five conversational events as regressors: turn initiation, production, comprehension, silence, and fixation (looking at a fixation cross in between conversations). In both datasets, the contrast turn initiation vs. fixation revealed engagement of classical language and sensorimotor areas. This contrast (along with the contrasts turn initiation > production and turn initiation > comprehension, which worked as controls for lower-level sensorimotor processing) also generated clusters of

activation in the midline structures of the frontal cortex, spanning from the medial superior frontal gyrus (mSFG) to the medial prefrontal cortex (mPFC; e.g., the anterior cingulate cortex). Based on these findings, we suggest that turn initiations are subserved by higher-order extra-linguistic processes, localized in the mSFG and the mPFC. These processes likely play a critical role in naturalistic language use, where speakers must continuously meet the temporal and pragmatic constraints governing timely and contextually appropriate turn-taking. Finally, we argue that adequate models of speech planning and large-scale neurobiological accounts of language must incorporate the interactive context in which language processing naturally develops and operates, namely, conversation.

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Adolescent pragmatic development mirrors pragmatic individual differences in adulthood: An fMRI study

by

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In conversation, the literal form of an utterance often differs from the speaker's intended meaning. Nonetheless, we typically understand the meaning of such indirect speech acts (ISAs). This constitutes one of several pragmatic abilities. fMRI studies in adults show that processing ISAs (compared to direct speech acts, DSAs) elicits greater activation in language

and Theory of Mind (ToM) regions. In our previous investigation, we found that individual differences in pragmatic skill manifested in two clusters outside these networks during ISA processing. However, less is known about the neurodevelopmental trajectory of ISA processing in adolescence. Using an established fMRI paradigm, we investigated ISA processing (1) across adolescence and (2) into adulthood by comparing adolescents to adults with varying pragmatic skill. We collected new fMRI data from adolescent participants (n = 51), divided into younger (ages 13—15, n = 24) and older (ages 16—18, n = 27) groups. Participants listened to short conversations ending with indirect or direct speech acts. We also included our previously collected adult data set (n = 57) in which participants completed the same paradigm. The adults were subdivided into two groups based on performance on pragmatic behavioral tests: a low-scoring group (LS, n = 28) and a high-scoring group (HS, n = 29). We contrasted activity during ISAs and DSAs within each group and tested for differences between groups, using whole-brain and ROI-based analyses. Our adolescent results replicated prior findings in adults. Activity increased between adolescence and adulthood in the posterior medial superior frontal gyrus (pmSFG). We also observed that activity in a cluster within the dorsal posterior cingulate cortex (dPCC), a core node of the Default Mode network, was higher during ISA in the HS group relative to all other groups (LS and adolescents). The increasing activity in the pmSFG suggests that this region may be a marker of pragmatic development, possibly reflecting a shift from general ToM to more conversation-specific pragmatic representations as conversational experience accumulates. The observed group differences within the dPCC indicate that pragmatic maturation may involve increasing reliance on internally directed processes, e.g. drawing on prior experience stored as situation models. Furthermore, the similarity between adolescents and the LS group indicates that a delayed development may lead to persistent difficulties in adulthood. Together, these findings suggest that pragmatic processing continues to develop through adolescence and that aspects of this development depend on neural systems beyond language and ToM.

Number-line training shapes short-term developmental pathways between approximate and symbolic numerical systems in preschoolers

by

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Understanding how children connect approximate nonsymbolic magnitudes with exact symbolic numbers is a central question in theories of numerical development. A major debate concerns the directionality of this relation: whether approximate magnitude representations scaffold the acquisition of symbolic number knowledge, as proposed by mapping accounts, or whether symbolic number learning refines approximate magnitude processing, as proposed by refinement accounts. The present study examined these bidirectional developmental pathways in Swedish preschoolers and, separately, whether number-line estimation training was associated with differences in their short-term predictive relations.

A total of 114 preschoolers participated in a pretest–posttest study with experimental and control groups. Children in the experimental group received three brief sessions of game-like number-line estimation training on a bounded 0–20 number line with trial-by-trial feedback, while the control group followed regular preschool activities. Symbolic and nonsymbolic magnitude comparison skills were assessed at pretest and posttest using the Numeracy Screener. To test the refinement pathway, posttest nonsymbolic performance was predicted from pretest symbolic performance, controlling for baseline nonsymbolic performance and condition. To test the mapping pathway, posttest symbolic performance was predicted from pretest nonsymbolic performance, controlling for baseline symbolic performance and condition. We then formally examined whether the nonsymbolic-to-symbolic predictive relation differed by condition using an interaction model.

Results showed evidence for a refinement pathway from SNS to ANS across groups: pretest symbolic abilities significantly predicted posttest nonsymbolic abilities after controlling for baseline nonsymbolic performance and condition. Conversely, pretest nonsymbolic abilities significantly predicted posttest symbolic abilities after controlling for baseline symbolic performance and condition, consistent with predictions from the mapping account. To test whether this mapping-related relation differed across groups, post-hoc analyses examined the interaction between pretest nonsymbolic performance and condition. The interaction was

positive and in the predicted direction, although it did not reach conventional two-sided significance. Complementary Bayesian sensitivity analyses provided stable directional support for a stronger nonsymbolic-to-symbolic slope in the experimental group, and group-specific regressions showed a significant relation in the experimental group but not in the control group. Taken together, these analyses provided converging, moderate directional evidence that the approximate-to-symbolic predictive relation was stronger in the experimental group.

These findings are consistent with a bidirectional view of numerical development in which symbolic knowledge predicts later approximate performance, while approximate numerical abilities predict later symbolic performance. They also suggest that number-line estimation training may be associated with a stronger approximate-to-symbolic predictive relation. By distinguishing evidence for developmental pathways from evidence concerning training-related differences between conditions, the study helps clarify mixed findings on the directionality of relations between approximate and symbolic numerical systems and highlights number-line estimation as a potentially useful activity for supporting early mathematical learning.

Session 4: Interaction, participation and space

Swedish driving examiners' communication methods during driving tests: Early findings from an ongoing field study

by

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Introduction

This study is part of an ongoing project, *Adjustment for Driver's License*, funded by the Swedish Transport Administration, which investigates how candidates requiring adjustments are treated during mandatory theory and driving tests. The project draws on observations and interviews to examine institutional practices related to support and adaptation in the driver licensing process.

Central to a driving test is communication, which constitutes a key cognitive skill enabling route planning, instruction comprehension, and coordination between examiner and candidate. Although test candidates are expected to carry out driving related actions independently, a driving test is a combined effort which, for example, requires *joint attention* (Kidwell & Zimmerman, 2007). To assess the test candidate's route-planning ability, the requested route must be established *intersubjectively* (Husserl, 1970), drawing on shared knowledge about the test candidate's support needs, among other things.

The Swedish Transport Administration's driving examiners welcome test candidates with diverse support needs including language difficulties and neurodevelopmental disorders. Work instructions state that driving examiners should be attentive to "different behaviour" and seek to understand candidates' language difficulties and "other special needs" (Swedish Transport Administration, 2018). However, information provided when a customer applies for a theory test with support is not transferred to the driving test, partly due to GDPR restrictions. As a result, examiners often lack prior knowledge of candidates' support needs and must identify suitable communicative methods *in situ*. On these grounds, this study sets out to explore driving examiners' communication methods during driving tests and the ways that these methods are adjusted to the test candidates' need for adaptations.

Aim and research questions

This study explores the communication methods used by driving examiners to convey crucial information during driving tests and how these methods are adapted to the test candidates'

needs. The following research questions will be pursued: *RQ1: What communication methods are used by driving examiners to convey crucial information to the test candidates during driving tests? RQ2: How are these methods adapted to the test candidates' needs? RQ3: How do driving examiners form an understanding of the test candidates' need for adaptations before the driving tests?*

Method

Data were derived from ethnomethodologically informed ethnographic fieldwork (Crabtree et al. 2000; Meier zu Verl & Meyer, 2024; Rouncefield, 2011; ten Have, 2004) conducted at two of the Swedish Transport Administration's test centres. Ethnomethodological research takes a participants' perspective as its analytical starting point, generating descriptions grounded in the social phenomena under investigation. Rather than relying on a priori theorising, it examines members' real-world practices as observable means through which social order is produced and sustained (Garfinkel, 1967). Participant observations of 60 driving tests were conducted. The data include jotted field notes and "headnotes" (Emerson et al. 1995); no audio or video was recorded (due to restrictions). Participants were recruited during a prior visit to the test centre. In total, nine driving examiners volunteered to participate, of whom one was female and eight were male.

Results

The findings are organized around the research questions and summarized at the end of this chapter.

Practices for passing on information and organizing attention during driving tests

Attention-organizing practices in driving tests are not categorically different from those found in driving lessons (cf. Björklund-Flärd, 2024). In the test environment, such practices are largely organized around navigational instructions, that is, directives that formulate directions for the ongoing driving. Consistent with the institutional orientation to independent driving, the observed tests did not include practices through which landmarks are jointly established and confirmed, practices that are routinely found in driving lessons (Björklund-Flärd, 2024).

A challenge that driving examiners may face is formulating navigational requests so that the test candidate can grasp them (and therefore be expected to comply with them). Such requests often involve the use of signposts with names of places or districts, which requires the ability to read Swedish. In situations where communication is impeded, for example due to language barriers, examiners may instead rely on road numbers or visual representations. Digital devices,

such as tablets, or traditional tools like pen and paper, may be used to spell out place names, produce schematic drawings, or present traffic symbols.

By contrast, driving lessons regularly involve correctives and other instructions, such as car-control instructions and monitoring instructions (De Stefani & Gazin, 2014). When comparable actions occur in the context of a driving test – for instance, when the examiner adjusts the steering by grabbing or touching the steering wheel, or verbally urges the candidate to brake for another vehicle that should have been noticed – these actions (referred to as "ingrip") are consequentially treated as grounds for failure. As part of the assessment process, examiners may ask candidates situational questions, such as: What were you thinking about the car there? (SWE: "Hur tänkte du om bilen där?"). These questions are designed to support the examiner's assessment and are not treated as consequential for the overall test outcome. These observations demonstrate the range of communicative resources used by driving examiners to convey directions and other information during driving tests, including verbal, visual, and material resources (RQ1).

Welcoming and preparing candidates for the test

The welcome phase is a central yet time-limited opportunity for identifying any potential support needs. Each driving test is assigned 45 minutes, and the driving should account for at least 25 minutes. The remaining time is used to accomplish preparatory and concluding routines and establish the test procedure.

During the initial rundown, driving examiners provide verbal and visual information about the test. Candidates who arrive with a driving school vehicle are typically briefed in their car, whereas private learner drivers, who do not bring a vehicle, may instead be received at the facility, where they are informed about the upcoming test. To visualize, driving examiners use a tablet displaying a tabbed interface with panels for introducing the test, checking of test points, and reporting the result (Figure 1).

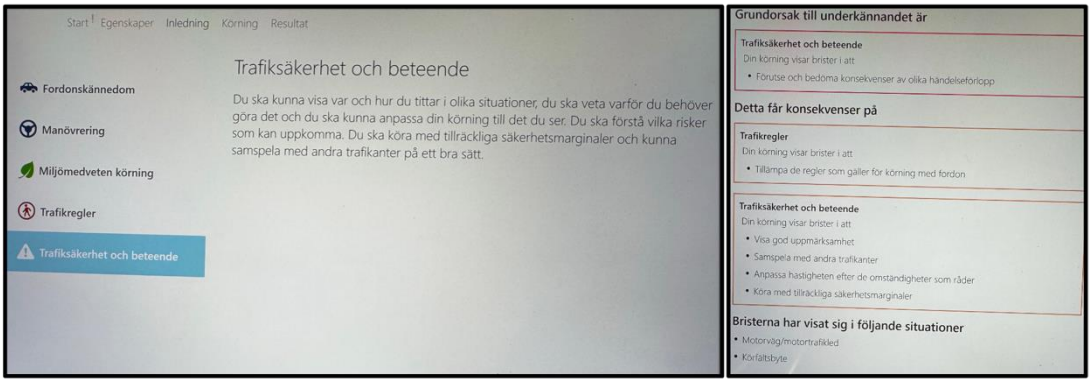


Figure 1: *An examiner's tablet*

The headers in the left column can be expanded to show more detailed information about each competency area. Some driving examiners leave them collapsed, allowing test candidates to focus on the oral presentation. Each area is often presented with concrete examples and examiners occasionally let test candidates give their own examples as a pedagogic feature. Test support videos in several languages (the use of which is encouraged) provide a resource when briefing candidates whose first language is not Swedish. In practice, these videos are rarely used; instead, examiners deploy their own, sometimes shifting, language skills when introducing the test to non-Swedish speakers. These examples show how examiners adapt their communicative practices to candidates' emerging needs by combining verbal, visual, and multilingual resources (RQ2).

Identifying candidates' support needs

In none of the observed tests were identifiable methods or practices employed that could be regarded as systematic strategies for eliciting information about the test candidate's specific needs. Instead, driving examiners rely on intuition to identify issues, such as test candidates' difficulties in understanding instructions before the driving test. "You get a feeling for someone really quickly when you say hello and start talking", one examiner expressed. During one observation, the candidate's failure to follow road signs was attributed to inadequate supervision rather than treated as a possible consequence of stress or cognitive impairment. Importantly, the test candidate's difficulties, regardless of their underlying cause, only became apparent while driving. These findings suggest that examiners primarily form an understanding of candidates' support needs through situated interactional judgments rather than through formalized procedures (RQ3). Table 1 presents a summary of the results, implications, and conclusions.

Table 1: *Summary of results, implications and conclusions*

Results	**Implications**
• Examiners flexibly draw on a combination of visual, verbal and embodied resources to inform test-takers and provide directions during driving tests. • To some extent, examiners appear to rely on intuitive judgments to identify issues, such as candidates' difficulties in understanding instructions during the driving test. • No identifiable methods or practices were employed that could be regarded as systematic strategies for eliciting information about the test candidate's specific needs in advance.	• Examiners may experience difficulties in acting fully in accordance with established work instructions, primarily due to time constraints and the absence of clearly defined routine descriptions. • When candidates' difficulties are identified only gradually during the assessment, certain candidates may be disadvantaged in the testing process. • Results underscore the importance of more structured approaches to identifying individual needs at an early stage, for example by incorporating open-ended questions at the outset of the test, aimed at eliciting relevant information about the candidates' specific needs (see Thorslund & Lidestam, 2025 for a discussion). • These findings suggest that a more equivalent testing process may be supported by welcoming candidates in a secluded area and by making greater use of translated test-support materials.
Conclusions	

This study demonstrates that communication in driving tests relies on a flexible combination of verbal, visual, embodied, and material resources. While driving examiners routinely adapt their practices to candidates' emerging needs, systematic procedures for identifying support needs prior to testing were not observed. The findings suggest that more structured approaches to eliciting relevant information at the outset of the test could support a more equivalent assessment process.

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Interpreting ‘Here’ And ‘There’ in shared Human-Robot Spaces

by

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For robots to interpret every-day language of users, they need to be able to disambiguate abstract terms. Spatial deixis, expressed through terms such as ‘here’ and ‘there’, is fundamental to how people refer to objects and locations in shared spaces. While these terms are typically described as proximal and distal expressions, their interpretation is not only affected by metric distance, but also by how a speaker conceptualizes the shared space [1]. This makes grounding spatial deixis essential in human-robot interaction (HRI), where a robot must identify the intended referent while also interpreting how the human conceptually frames the shared space.

Linguistically, spatial deixis varies across languages. English uses a binary distinction, while languages such as Spanish and Japanese use three-way distinctions [1-4]. Neuropsychological research maps this distinction to the visuo-tactile space, namely the peri-personal or reachable space and extra-personal space [5–7].

Motivated by this cognitive perspective, we proposed a Spatial Deixis Model (SDM) [8] that enables robots to ground and interpret English spatial deixis. In a preliminary HRI experiment, participants interacted with a Pepper robot in a shared cupboard environment containing cups and bowls, arranged across five scenarios with different spatial configurations and object density. In each trial, participants pointed to objects while referring to them using expression containing ‘here’ or ‘there.’ The robot detected the pointing gesture, localized the intended object, and predicted a spatial deixis with respect to the person. For every pointed object, we recorded the participant’s verbal deixis and the model’s predicted SDM label.

Across 174 (out of 180) valid trails with 12 participants across all five scenarios, the robot localized the pointed-at object with 92% accuracy. However, agreement between participants’ verbal deixis and the model’s pointing-based labels was only 63.7%, with scenario-level agreement varying between 50% and 72.2%. This gap, and its variation across scenarios, revealed a clear inconsistency in how participants used spatial deixis in a multi-object setting. This led us to consider that participants’ linguistic choices might be shaped by factors such as environmental context (quantity and arrangement of objects), the relative closeness of objects, and the human’s embodied perspective. From these factors, we constructed five candidate mental models. The Absolute model (AM) is driven by the contextual environment, where the

cupboard's spatial layout provides a structured frame biasing deixis selection. The Relative Model (RM) considers an object's position relative to other objects. The Mixed Model (MM) integrates both absolute and relative interpretation. The Temporal model (TM) captures carry-over effects, examining whether deixis assigned to an object in one arrangement is influenced by its interpretation in preceding arrangements, reflecting mental conditioning or adaptability across scenarios. The Response-biased model (RbM) applied to participants who consistently favor one deixis term across most objects. Results showed that AM was more prominent in simpler configurations, while complex scenarios involved combinations of models.

These findings suggest that robots should incorporate multiple interpretive strategies to capture variability in human spatial deixis. Future work will evaluate SDM on other spaces and more participants to generalize its conceptuality.

ACKNOWLEDGEMENTS:

Adwitiya Mandal's work is funded by the Swedish Research Council (VR) under grant 2022-03966. Zoe Falomir acknowledges the funding by the Wallenberg AI, Autonomous Systems and Software Program (WASP) awarded by the Knut and Alice Wallenberg Foundation, Sweden.

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Relational Epistemology of Biological Agency with Artifactual Scaffold

by

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Abstract

Within relational epistemology of computing nature, knowledge is generated through the interactions of embedded agents on different levels of organization. This results in the observer's ladder, where the "observer" is a participatory node in a network of info-computational agents. From the simplest organisms to humans, living beings are cognitive agents. In the interaction with the environment, many of them use tools to extend cognitive processing. This understanding of the evolution of cognitive agency helps frame contemporary GenAI as an artifactual scaffold of the human mind.

Keywords: Relational epistemology, observer-agency, info-computationalism, morphological computing, cognition, agency, extended mind, AI

Agency and Extended Cognition

The argument is developed based on the ICON (Information-Computation-Cognition Naturalism) framework presented in *Evolving Intelligence* (see the book overview, Dodig-

Crnkovic, 2026). Starting with the concept of the minimal observer, it distinguishes three levels of cognition in nature: biological cognition as capacity of all living organisms, human cognition extended by tools including AI, and hypothetical hybrid systems consisting of humans in interaction with AI-agents.

Organizational Transitions in Biological Agency: The Observer and the Ladder

Relational epistemology describes knowledge as a result of interactions between situated observer-agents and their environments (Dodig-Crnkovic, 2025). A minimal observer-agent in this framework registers differences in the world through changes in its own state and affects other systems (acts). Agents include natural physical, biological, and artificial systems. Information is a difference that makes a difference for a particular system (Bateson, 1972). Computation is a change in informational structure through physical processes of morphological transformation (Dodig-Crnkovic, 2017).

The *observer's ladder* describes new capacities as arising when existing processes become integrated into new forms of organization. Physical and chemical interactions provide conditions for living self-maintenance. Living cognition supports symbolic practices that externalize memory. These symbolic practices supply materials and purposes for artificial intelligence through dependencies with feedback.

The threshold defining biological cognition is autopoiesis: a causal organization through which a living system continually produces and maintains itself (Maturana & Varela, 1980). Cognition develops through the organism's interactions with its environment, with causal relations becoming meaningful relative to its viability and organization. Adaptive information processing is integrated into this ongoing activity: environmental changes matter through their consequences for the organism's continued existence. Biological agency is the capacity to regulate these interactions in relation to viability. Its distinctive feature is how causal interaction is organized around living self-maintenance. Intelligence consists in flexible problem-solving capacity, which can also be realized artificially.

From Biological Cognition to Artifactual Scaffolding

Language, writing, instruments, and scientific models allow living agents to organize inquiry through external resources. This provides the bridge from autopoiesis to extended cognition: the processes supporting a cognitive activity exceed the organism's boundary without every participating component becoming autopoietic. This scaffold represents a rhizomatic extension (Deleuze & Guattari, 1987) resulting in a new, hybrid mutualistic cognitive architecture (Clark,

2008). It reorganizes the causal interactions through which living agents inquire and act. Clark (2025) situates generative AI within this history of cognitive extension.

AI can transform externalized knowledge and alter the possibilities available to a human inquirer. Its contribution is therefore causally consequential, even though it lacks biological agency. The relevant comparison concerns both the source of goals and the organization of interaction.

The Spectrum of AI Agency in Scientific Inquiry

When a scientist develops an explanation through dialogue with generative AI, where does agency reside? Until recently, AI has been used as a computational tool. Its rapid development toward autonomous agency, however, is altering this relationship and may lead to a hypothetical hybrid system in which human cognitive agency is coupled to AI's intelligent agency. This paper argues that cognitive activity can extend across a human-AI system while the sources and goals of cognitive agency remain human.

As an example, consider a scientist studying a new experimental result. The following three cases are conceptual illustrations. In the first, the scientist uses a statistical program to analyze experimental results. The program transforms data within a procedure selected by the scientist. Intent resides in the scientist's purpose of explaining the result. Human biological cognition motivates and organizes the inquiry, and the software provides computational resources. This is an example of cognition extended by software.

In the second, the scientist asks a language model for alternative explanations and revises the conclusions in response. The model contributes problem-solving proposals and can, for example, direct the scientist's attention toward an overlooked assumption. Its outputs influence the inquiry, but problem definition and goals are provided by humans. Problem-solving, reasoning, efficient search across vast data and knowledge databases, and conversational fluency do not, by themselves, establish cognition or autonomous intent in an LLM. The scientist frames questions and evaluates proposals, while the model's behaviour reflects objectives and constraints set by researchers, developers and institutions.

In the third case, a hypothetical AI system retains the project's history, uses experimental feedback, and updates proposals across successive investigations. The scientist's strategies and the system's responses become mutually adjusted over time. Persistent memory and reciprocal adjustment make this a stronger candidate for an integrated (hybrid) cognitive system. Such integration could be investigated by examining how removing the scaffold changes hypothesis formation and learning.

Across these cases, information processing becomes distributed while living self-maintenance and meaningful goal-directedness remain with the human researchers. The observer's ladder distinguishes an extension of an existing observer's capacities from the emergence of a new organizational form of agency.

Conclusion

The observer's ladder contributes an organizational account of how cognitive scaffolds become possible and a way to distinguish component agency from the activity of a coupled system. Its application here connects the generative project of *Evolving Intelligence* to the analysis of human-AI inquiry. The extension of cognitive activity and the biological grounding of cognition are compatible. Assessing the nature of a particular human-AI configuration requires examining its goals, feedback, and integration.

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No credible evidence that education protects cognition against brain aging: Converging evidence from causal simulations and longitudinal neuroimaging studies

by

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Abstract

Prominent theories in cognitive aging hold that education builds *cognitive reserve*—an adaptive brain characteristic theorized to attenuate the adverse effects of brain aging and neuropathology on late-life cognition. According to this influential view, education acts as a protective factor that preserves cognitive performance in old age and lowers dementia risk. In this talk, I present dissertation work that challenges this account on both methodological and empirical grounds. Using causal modeling and Monte Carlo simulations, I first show that the cross-sectional evidence cited in support of the theory cannot identify the attenuation mechanism it predicts, and that statistical collider bias can plausibly account for substantial portions of the cross-sectional findings in the cognitive reserve literature. I then present converging longitudinal findings from three studies spanning multiple neuroimaging modalities in healthy older adults. The first uses harmonized data from the Lifebrain project (n = 708; mean baseline age 67.9) to test whether education moderates the coupling between change in hippocampal volume and change in episodic memory. The second and third use the Cognition, Brain, and Aging (COBRA) cohort (n = 181; baseline age 64–68), in which participants underwent repeated dopamine PET ([11C]raclopride), structural MRI, and cognitive assessment, to test whether education moderates couplings between change in striatal D2-like receptor availability and change in global cognition, working memory, episodic memory, and processing speed over five- and ten-year periods. Ongoing work extends these tests to gray-matter measures in COBRA. Across more than 40 statistical models, I find no credible evidence that education attenuates brain–cognition coupling in healthy older adults (Lövdén et al., 2023; Pagin et al., 2026a, 2026b). Overall, these results are difficult to reconcile with cognitive reserve theory, but align with other recent longitudinal work on education in neurocognitive aging. The emerging picture is that education predicts late-life cognitive performance largely because it proxies early-emerging individual differences in cognitive ability that remain stable across the lifespan.

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Session 5: Olfaction, semantics and aging

A data-driven approach for mapping the semantic organization of smell

by

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Odor vocabularies of western languages are underdeveloped in comparison to those of other sensory domains (e.g., Majid 2021). There are almost no terms that only are used to describe odor experiences (such as, for example, color words). Odors are instead described e.g. by reference to the source (*buttery*), intensity (*overpowering*), familiarity (*unmistakable*) or the hedonic evaluation (*pleasant*) of the odor, or with cross-modal or other metaphors (e.g., *heavy*) (e.g. Hörberg et al. 2022; Poulton 2020). Odor vocabularies of western languages are therefore vague and unspecific, making it difficult to delineate the words people actually use to describe odors.

In this talk, we will present a data-driven method that finds the words people actually use to describe odors in natural language, and then maps the odor-perceptual or odor-semantic organization of those words. This method uses a natural language processing (NLP) approach that automatically identifies odor descriptors in natural texts and then derives the semantic relationships between those descriptors using (large) language modeling ((L)LM) (Hörberg et al. 2022).

The quality of this semantic organization will depend on the internal representations of the (L)LM used. We evaluate the representations of three generations of (L)LMs, including under nearly 200 training configurations with different training data (Kurfalı et al. 2025). We benchmark the model representations and their resulting odor-semantic spaces against odor similarity ratings, capturing odor-perceptual relationships, as well as odor label similarity ratings and odor-to-label ratings, representing odor-semantic relationships. Simpler models best capture odor-perceptual similarities, while state-of-the-art LLMs excels in simulating odor-semantic relationships. The quality of the model's odor-perceptual and odor-semantic representations is further highly dependent on the quality of the training data, with better representations of models trained on odor-related texts.

We provide a data-driven approach for deriving olfactory vocabularies. Our findings show that natural language encodes latent information about human odor knowledge that is retrievable through text based (L)LMs to some degree, and they highlight the potential of using (L)LMs to map odor-perceptual and odor-semantic spaces, using our method.

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Free odor identification as a tool to study the aging mind

by

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Correctly naming household odors such as thyme, cardamom or strawberry can be challenging, even for individuals with a keen sense of smell. Due to this difficulty, researchers typically use multiple-choice tests of cued odor identification (OID) instead of free odor identification when olfaction is assessed. Cognitive, neuropsychology, and aging researchers have found that cued OID impairment is a strong risk factor for incident dementia. However, little is known about the cognitive aspects of free OID. In this study (Hörberg et al., 2025), we analyze the demographic and cognitive factors that influence OID responses of old adults. We utilize a uniquely large dataset (n = 2,479) from a population-based sample of healthy, older Swedish

adults (ages 58–102) who participated in free and cued OID using the 16-item Sniffin' TOM test. The free OID naming responses were categorized as correct, misnamings, or omissions. The results revealed that omissions are surprisingly prevalent, constituting 66.4% of errors and accounting for 87.7% of the age-related differences in task performance. Additionally, we hypothesized that successful free OID would be more closely linked to nonolfactory cognitive abilities, such as verbal fluency, vocabulary, and episodic memory proficiency. This hypothesis was supported, as we found significant associations between free OID and these cognitive abilities, while cued OID identification only was associated with perceptual speed. Our findings suggest that the assessment of free OID may provide valuable insights into odor-based cognition. I will outline how free OID might predict the risk of incident dementia in older adults, beyond what is possible with typical olfactory assessment.

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Sex Differences and Female Hormonal Factors in Olfactory Decline and Dementia: A 12-Year Longitudinal Study

by

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Abstract

Women bear a disproportionate share of the dementia burden, yet the hormonal mechanisms underlying this disparity remain incompletely understood. Hormonal factors could also influence olfactory performance. Estrogen loss during the menopausal transition has been linked to reduced neurogenesis in the dentate gyrus, impaired maintenance of olfactory sensory neurons, and measurable declines in olfactory function (Ekanayake et al., 2024; Mervosh & Devi, 2025; Singh et al., 2019). Reproductive span, defined as the interval between menarche and menopause, serves as a proxy for cumulative endogenous estrogen exposure across the lifespan (Georgakis et al., 2016). However, longitudinal evidence specifically linking reproductive history to the trajectories of olfactory aging remains scarce.

Olfactory identification (OID), the ability to recognize and name familiar odors, extends beyond basic sensory processing and relies on memory retrieval, semantic access, and executive function. Consequently, OID represents a sensitive marker of cognitive aging. Impaired OID performance has been shown to predict future dementia before clinical diagnoses, highlighting its potential as a low-cost, non-invasive tool for identifying individuals at elevated neurodegenerative disease risk (Laukka et al., 2023; Olofsson et al., 2021).

This study examined sex differences, reproductive span, hormone therapy (HT) use, and *APOE* ε4 carrier status as predictors of longitudinal OID trajectories and incident all-cause dementia over 12 years in a population-based Swedish cohort (SNAC-K; N = 2,479, 61.3% female). It was hypothesized that longer reproductive span and prevalent HT use would be associated with slower OID decline, and that lower OID scores would predict incident dementia.

Linear mixed-effects models (LMMs) showed significant OID decline over the follow-up period, with women showing consistently outperforming men. Crucially, women with longer reproductive spans showed a significantly slower rate of OID decline (time × reproductive span interaction, p = .037). Specifically, each standard deviation increase in reproductive span (~3.23 years) was associated with approximately 0.36 additional OID points preserved over 12 years. This finding supports the proposed neuroprotective role of estrogen and suggests that cumulative endogenous estrogen exposure contributes to individual differences olfactory aging trajectories. *APOE* ε4 carrier status was associated with accelerated olfactory decline in the full sample. Although HT use was associated with better OID performance at baseline, it did not

independently predict the rate of subsequent decline. Cox proportional hazards regression further indicated that each one-point increase in baseline OID score was associated with a 9% reduction in dementia hazard (HR = 0.91). This relationship remained significant after adjustment for vascular risk, stroke, reproductive history, HT use, and *APOE* ε4 carrier status.

Together, these findings identify reproductive history as a important source of variability in olfactory cognitive aging and reinforce the value of OID as a sensitive marker of neurodegeneration. Women with shorter reproductive span or known *APOE* ε4 carrier status may represent a priority population for such screening, given their increased vulnerability and potential to benefit most from early detection and intervention.

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Session 6: Memory, perception and prediction

Prediction errors shape episodic memories: Empirical evidence and a novel framework

by

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The brain is thought to constantly generate predictions about upcoming sensory input. If these predictions are incorrect, a prediction error (PE) is generated that can act as a learning signal to update internal models (Friston & Kiebel, 2009). Recent research has found that episodic memories can be leveraged for predictions and that PEs can trigger new episodic encoding but also modification of existing memories (Sinclair & Barense, 2018). However, there has been mixed evidence concerning which behavioural outcomes materialize after PEs and how they are implemented representationally and neurally (Bein et al., 2023). Consequently, there has been no theoretical framework in the field that can account for different episodic memory outcomes after PEs.

In three studies, two of them using fMRI, we found strong evidence that the size of the PE is decisive in shaping episodic memory outcomes. In all studies, episodic memories of naturalistic conversations were induced in participants. Later, the stimuli were re-played, but with an unpredicted change, aimed to evoke a PE. PE size was manipulated by employing different magnitudes of change and measuring the strength of the prediction. Recognition memory was tested with a delay of at least one day.

Across our studies (Boeltzig et al., 2026; Boeltzig, Liedtke, & Schubotz, 2025; Boeltzig, Liedtke, Siestrup, et al., 2025; Liedtke et al., 2025; Pronoza et al., 2026), small PEs lead to little or no new encoding, as predictions are largely correct. Moderate PEs led to updating of the existing memory, characterised by representational change and a neural profile of integration (vmPFC, BA9) and updating (TPJ, IFG), leading to reduced recognition memory. Large PEs triggered separate encoding of the unpredicted event, leaving the original memory trace representationally intact, and yielding high recognition memory for original (i.e., predicted) and modified (i.e., PE-evoking) versions of the event. Correlated with these effects, and therefore a candidate for an underlying mechanism triggering this separate encoding, may be increased reinstatement of the original memory trace in the face of the incoming mismatch.

These results are leveraged in the novel REPRISE framework, which aims to account for diverging effects of PEs on episodic memories in prior studies. It places PE size at its centre and argues that it shapes representational and behavioural outcomes, in line with the described effects. PE size can exert its influence both with the aid of relevant brain regions and via increased original reinstatement. REPRISE can therefore unify previous experimental results, account for null findings in prior studies, and form the basis for future research questions and experimental designs within the field.

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Reassessing Unconscious Visual Working Memory: Evidence from Contralateral Delay Activity

by

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Visual working memory (VWM) allows us to maintain visual information to guide behavior. It is traditionally considered closely linked to consciousness, with VWM representations assumed to be consciously accessible. However, behavioral studies suggest that VWM can persist even when participants report no awareness of the stimuli. Further, a recent study examined the question of unconscious VWM using tilted Gabor patches and the contralateral delay activity (CDA), an electroencephalographic measure of VWM. Findings suggested that when participants reported awareness of something, but not the orientation, CDA amplitude was larger on correct than incorrect trials. This was interpreted as evidence for unconscious VWM. However, this study (N = 14) used a small sample and lacked validity checks of awareness ratings. In the present study, participants (N = 40) viewed lateralized, backward-masked Gabor patches to elicit CDA. After each trial, they rated their awareness and attempted to reproduce the orientation. A continuous staircase procedure adjusted contrast to individual awareness thresholds. As behavioral validity checks, the task also included empty catch trials and clearly visible control trials. In Bayesian linear mixed models, CDA was modelled in terms of performance (absolute orientation error), awareness, Gabor contrast, and whether participants fulfilled all behavioral validity checks (20 did and 20 did not). Results suggested that when participants reported unawareness (seeing only something), absolute orientation error did not predict CDA (BF01 = 383). This finding weakens the claim that CDA is tied to performance even if participants report unawareness.

Patterns of sensory misattribution in spatial memory: A preregistered virtual reality study

by

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New memories often interfere with old memories, and retroactive memory interference effects are well-established within sensory systems. Although people perceive the world through several senses, interference across senses is not well understood. Recent reviews emphasize that spatial memory may differ across the senses, and theoretical accounts propose that the human spatial navigation system co-evolved with olfaction, suggesting a prioritized role for olfaction in spatial memory. In our previous study, we found preliminary evidence that olfactory dominance in retroactive interference might be reflected in the specific patterns of memory errors participants make. Specifically, when making errors, participants tended to place sounds in location of concept-matched smells rather than vice versa. This particular type of error, sensory misattribution, is the subject of our ongoing study. In a preregistered experiment, we aimed to (1) replicate previous results indicating asymmetry in sensory misattribution favoring olfaction, and (2) test whether some types of odors are more likely to elicit sensory misattribution asymmetry. To that end, we selected objects of different levels of biological relevance, smells and sounds related to and not related to food. Participants memorized locations of concept-matched smells and sounds, presented in a blocked design, within a Virtual Reality arena. After encoding locations in each sensory condition, participants recalled these locations in an immediate recall session (T1) and later, after a distraction task, at a delayed mixed modality recall session (T2). The arena was a green meadow surrounded by a circular wall and visual landmarks for spatial orientation. Half of the objects were edible (coffee, bacon, coca-cola, popcorn), and half were not (grass, old book, sawdust, laundry). Concept-matched smells and sounds were always at distinct locations (e.g., smell of popcorn was at a different location than sound of popcorn). Data analysis is currently ongoing, but preliminary results based on the Bayesian analyses with default priors support the presence of sensory misattribution asymmetry favoring olfaction, but the asymmetry manifested in proactive rather than retroactive interference effects. These preliminary results are in line with the prioritized role of olfaction in cross-modal spatial memory interference, however

biologically relevant odors do not seem to amplify this effect. Nevertheless, odor memories seem to have a disruptive influence on sound memories that share a common source, and olfaction might be prioritized in human spatial memory. After final analyses, which include specifying priors informed by the previous research, our findings will contribute to a deeper understanding of how different senses interact when forming and retrieving spatial representations. Result will be presented on Swecog2026

Session 7: Context, interpretation and the social mind

A different language, a different self? How second language use shapes risk aversion and personality traits

by

Oleksandra Osypenko (*Lund University, Sweden*) and **Alexandra S. Dylman** (*Mälardalen University, Sweden*)

Three lines of research have developed largely independently in the study of decision-making. First, research on the Foreign Language Effect (FLE) shows that individuals tend to be less risk-averse when making decisions in a foreign language compared to their native language (Costa et al., 2014; Del Maschio et al., 2022). Second, individual differences in personality traits have also been linked to variation in risk-taking behaviour (e.g., Joseph & Zhang, 2021; Nicholson et al., 2005). Third, emerging evidence suggests that personality traits themselves may vary across language contexts (Chen & Bond, 2010; Dylman & Zakrisson, 2023). However, little is known about how language context (i.e., the language in which the task is completed), personality traits, and bilingual experience interact to shape risk-related decision-making in bilinguals. The present study addresses this gap by examining the combined and interactive effects of language context and personality traits on risk aversion in bilinguals.

Across two experiments, Swedish–English (n = 100) and Italian–English bilinguals (n = 81) living in the UK completed a Big Five personality inventory (BFI-2-XS; Soto & John, 2017) and a visual lottery-choice task measuring risk aversion (Vasco & Francisco, 2025) in both their native language (L1) and a second language (English; L2). A brief distractor task was included between language blocks to facilitate a shift in language mode. We predicted reduced risk aversion in the L2 and that personality traits linked to exploration and reward sensitivity (e.g., extraversion, openness) would be associated with greater risk-taking, whereas traits linked to caution (e.g., conscientiousness, neuroticism) would be associated with greater risk aversion, particularly in the L2 context. We also examined how personality traits vary within participants across language contexts.

Results revealed a robust FLE in both experiments: participants were consistently less risk-averse in the L2, as indicated by earlier switching to risky options and a higher proportion of risky choices. However, the role of personality traits in risk aversion was more limited and emerged primarily in the L2 context. Specifically, L2 neuroticism was associated with greater

risk aversion in Swedish–English bilinguals, whereas in Italian–English bilinguals' L2 extraversion was linked to greater risk-taking and openness to greater risk aversion.

Importantly, across both experiments, language context also influenced personality reports. While some trait differences were observed across groups in the L1, consistent with previously reported cross-cultural variation (Kajonius & Giolla, 2017; Schmitt et al., 2007), these differences were reduced in the L2. That is, Swedish–English and Italian–English bilinguals showed more similar personality profiles when responding in English than in their native languages. For example, traits such as conscientiousness and neuroticism, which differed significantly across groups in the L1, showed more similar levels across groups in the L2, possibly showing awareness of the L2 cultural norms; yet this needs to be examined further. Together, the findings indicate that the FLE extends beyond risk preferences to include personality reports, and that language context can promote convergence in both decision-making under uncertainty and self-perception across culturally distinct groups.

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Gaze Allocation During Dynamic Emotion Recognition: An Eye-Tracking Study of Single and Blended Facial Expressions

by

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Facial emotion recognition is often studied using static photographs, even though natural expressions unfold dynamically and may communicate several emotional signals at once. We investigated how visual attention is organised during recognition of dynamic facial expressions, and whether looking at theoretically diagnostic facial regions actually supports successful recognition.

Fifty-three adults completed two eye-tracking studies. In Study 1, participants viewed 78 videos depicting 13 single emotions at two intended intensity levels. Study 2 used 60 videos depicting pairwise blends of anger, disgust, fear, happiness, and sadness. Gaze was recorded with a Tobii Pro Fusion eye tracker and mapped onto dynamic areas of interests for the eyes, nose, and mouth. The primary analyses examined total fixation duration and time to first fixation, together with recognition performance and regional facial action-unit activity.

Across both studies, emotional content systematically organised where and when observers looked. In Study 1, gaze allocation varied across emotion, facial region, and, for some contrasts, intended intensity. In Study 2, fixation duration and first-fixation timing likewise differed across intended blends, with perceived clarity modifying some regional gaze contrasts. However, this structured gaze did not translate into a simple recognition advantage. Neither longer fixation on, nor earlier access to, literature-defined diagnostic regions reliably predicted recognition accuracy in Study 1 or detection of individual blend components in Study 2. For regional facial movements, where significant, stronger action-unit activity was associated with

reduced or later gaze to the corresponding region rather than increased or earlier gaze. Cross-study comparisons further showed that the same emotions were sampled differently when presented alone versus within blends, including a relative shift towards the eyes and away from the mouth in blended expressions.

Our findings therefore contribute to the existing literature by distinguishing gaze allocation from information use. The results also provide support a more flexible account of dynamic emotion perception in which diagnosticity is context-sensitive and thus changes with the temporal and emotional structure of the display. In other word, contrary to the notion of a fixed "look at the right region" strategy, successful recognition appears to depend on distributed sampling of evolving facial information, with overt gaze providing only a partial index of the information used for categorisation.

The left temporo-parietal junction supports context-based ambiguity resolution

by

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Keywords: quantifier scope ambiguity, contextual language processing, pragmatics, perisylvian language network, LTPJ, fMRI

A central feature of language comprehension is the ability to resolve linguistic ambiguity, i.e. cases in which a linguistic unit can be interpreted in more than one way. The neural mechanisms underlying this process remain insufficiently understood, particularly with respect to different forms of ambiguity. This study focuses on quantifier scope ambiguity (QSA), which arises in sentences such as every child climbed a tree. Such sentences permit both singular-referent interpretation, in which all children climbed the same tree, and a plural-referent interpretation, in which each child climbed a different tree. Behavioral evidence suggests that listeners preferentially adopt the plural interpretation in the absence of contextual cues [7, 8]. However, preceding discourse context can bias interpretations [8–10]. Semantic processing has traditionally been associated with a left lateralized language network, particularly the left posterior middle temporal gyrus (pMTG) and the posterior superior temporal sulcus, which are

thought to support lexical and combinatorial semantic operations [6]. However, resolving QSA may require neural processes that extend beyond core lexico-semantic computations, as listeners must integrate discourse level information to constrain interpretation. The present study investigated this possibility using functional magnetic resonance imaging (fMRI). Thirty-two native Swedish speakers listened to a series of trials. The critical trials consisted of two utterances, one contextual sentence and a QSA sentence. Contextual sentences prompted either a plural or a singular interpretation. After each trial, participants indicated their interpretation via button press. A third of included trials included no QSA and served as distractors. Functional imaging data were analyzed in SPM using a general linear model including regressors for the context utterance, QSA utterance, question prompt, response period, and fixation. All events except fixation were modeled separately for singular, plural, and distractor conditions. The contrast comparing singular-prompted relative to plural-prompted QSA utterances revealed a predominantly left lateralized network extending from the anterior temporal lobe into the middle temporal gyrus, superior temporal gyrus, temporo parietal junction (TPJ), and inferior parietal cortex. The involvement of the left TPJ is particularly noteworthy, as this region is not typically considered part of the core language network, but has instead been linked to higher-order inferential processes, including belief attribution and perspective-taking [1, 3]. The left TPJ has also previously been implicated in homonym selection during word production [2]. This activation was observed during singular-prompted QSA only, not for plural prompts, which suggests that identical QSA sentences are processed differently depending on contextual information. It also implies that a less frequent interpretation of the grammatical structure of a sentence recruits additional activations, compatible with a theoretical notion of grammar as a tool for predictive processing [4]. These findings may suggest that certain contextual disambiguation recruits neural mechanisms extending beyond canonical temporal language regions.The results can also support recent neurolinguistic accounts proposing that regions at the margins of the language network [5] play a crucial role in naturalistic language comprehension.

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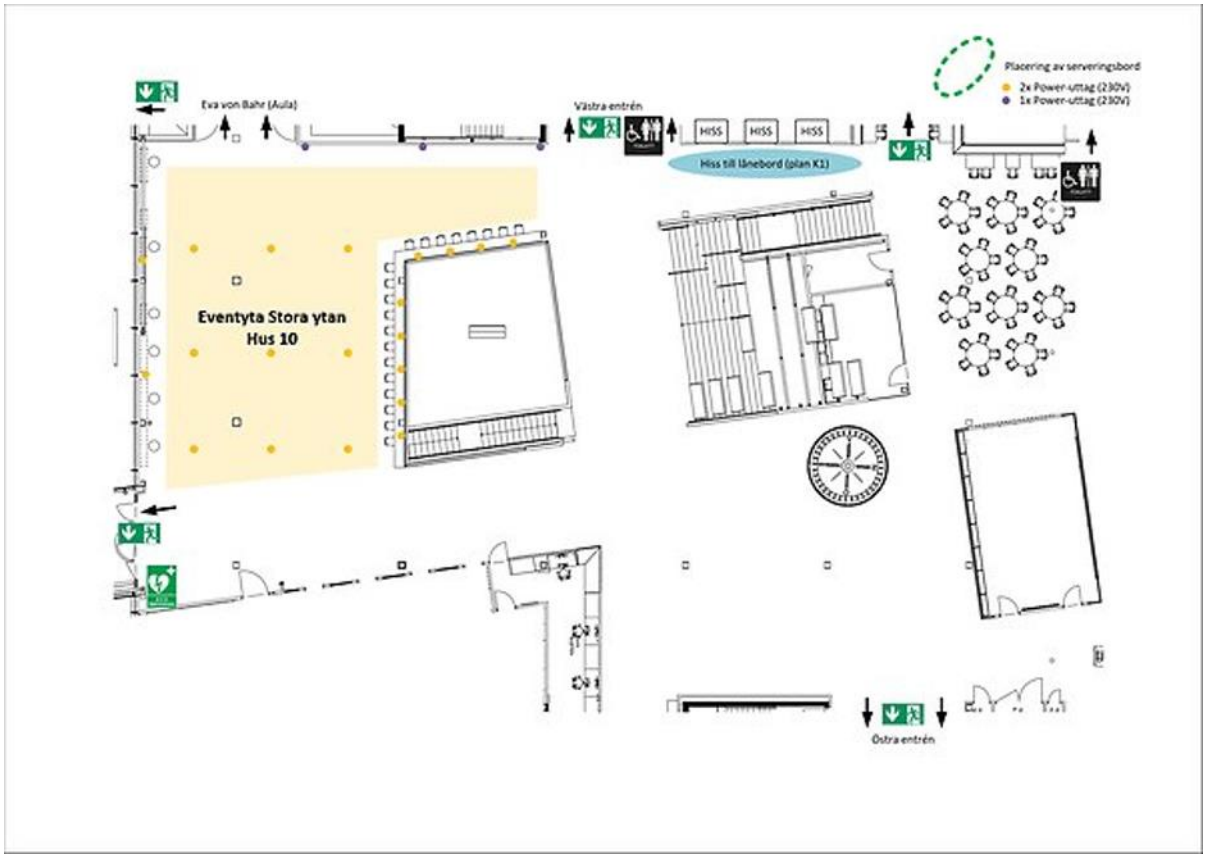
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Poster sessions with short papers and abstracts

The poster sessions are at the entrance area, floor 0, House 10, Ångström



Short papers

Rating Cognitive Process Involvement in Hazard Perception at the Swedish Driving Theory Test

by

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Introduction

Within traffic safety research and practice, it is crucial that assessments of new drivers effectively target the most important driving skills. Novice drivers have higher crash rates than experienced drivers, largely due to their limited driving skills and a greater tendency toward risk-taking behavior (Van der Kint et al., 2024). Hazard perception has shown to be a key driving skill that distinguishes drivers involved in traffic accidents from those who are not (Horswill, 2016). In Sweden, hazard perception is currently assessed as part of the theoretical driving exam, in which candidates are shown a single image or short video and must answer a multiple-choice question based on it. According to Barragan et al. (2021), several definitions and theories of hazard perception have been proposed. This paper focuses on the Hazard Perception-Response Framework suggested by Barragan et al. (2021) because it explains the cognitive processes underlying the detection of and response to hazards while driving. Based on this framework, hazard perception involves a wide range of cognitive processes (Barragan et al., 2021). However, it remains unclear which specific processes are represented in the hazard perception component of the Swedish theoretical driving test, as the test items are not currently designed based on a cognitive framework. This paper contributes to cognitive science and traffic research by being the first study to evaluate the videos used in the Swedish theoretical driving exam from a cognitive science perspective. It reports the results from five experts' ratings of the driving scenarios currently used in the driving test, focusing on their involvement of cognitive processes. The findings offer an initial understanding of how, and to what extent, hazard perception test material engages cognitive processes, while highlighting the methodological challenges that future test design must address.

Aim and Research Questions

This study aims to investigate which cognitive processes can be classified in the current theoretical part of the Swedish driver test by raters. As a result, instead of focusing on the

cognitive processes involved in real-world driving (e.g., motor coordination), the focus was placed on those that can be assessed using video-based testing. In addition, the raters decided to rate auditory processing as not involved (level 0) in all situations because all the videos are mute. The research questions are: *What cognitive processes can be rated in the Swedish driving test material according to cognitive scientists? What is the perceived level of involvement of each process in relation to the other videos as rated by experts?*

Method

Initially, a list of 29 cognitive processes required for hazard perception in safe driving was compiled from previous literature (Barragan et al., 2021; Moran et al., 2020). The list of processes and their definitions was provided to the raters while coding. To determine which cognitive processes can be identified as being involved in the video-based test material, five experts in cognitive science rated the level of involvement from 0 (no involvement) to 5 (high involvement) of a specific cognitive process in the media (27 videos). The study used data from the Swedish Transport Administration, which consisted of question ID, media name, the test question without answer options, the passing rate for each question, and the test category (driving license Type B). Additionally, 27 videos with matching media names in Excel were provided. Videos have a duration of 4 to 9 seconds, consisting of various traffic environments like following a city bus or driving on country roads, in a range of weather conditions. The rating was conducted by five raters independently by watching each video an average of 3 to 5 times and reading the associated question on a pre-defined screen size in an online tool. They could revisit the definition of each process until they were certain of the level of involvement for each process. These ratings represent expert judgments about the cognitive processes likely to be involved in each item, rather than direct measurements of the processes actually engaged during task performance. It remains possible that test-takers experience cognitive processes not identified by the raters or rely on the identified processes to a different degree than rated. To address RQ2, the videos with the highest levels of rater agreement (with a maximum agreement of 40%) were selected, and average involvement ratings were calculated for these videos to enable comparison across scenarios.

Results

Regarding the first research question, a wide range of cognitive processes were considered as being involved in the videos, though inter-rater agreement was relatively low. Among the 27

video scenarios, 10 showed the highest levels of agreement between raters (30% and 40%), whereas 17 showed agreement of 10% or less regarding the cognitive processes involved. The driving situations were perceived by raters as involving different levels of cognitive processing. For example, in the situation depicting driving on a snowy country road with two vehicles following behind, raters rated social cognition as having low involvement (M = 1.2 out of 5), whereas divided attention/multitasking received the highest level of agreement among raters, with an average involvement rating of 3.8 out of 5.

For the second research question, three video scenarios with 40% agreement among raters are shown in Figure 1: Roadworks and Lane Closure, Passing Car Near Highway Exit, and Bus Stop with Crossing Pedestrians. Raters assigned similar ratings to spatial processing. Despite different traffic contexts, each scenario required drivers to maintain awareness of vehicle position and surrounding traffic while anticipating upcoming environmental changes. These similar ratings suggest that raters shared a common perception of the extent to which spatial processing was involved across these scenarios. However, raters noted that the Bus Stop with Crossing Pedestrians scenario demanded more social cognition compared to the other two videos. The presence of pedestrians near the road likely heightened the need for drivers to interpret and anticipate the intentions and behavior of others, such as judging whether a pedestrian was about to step into the road.

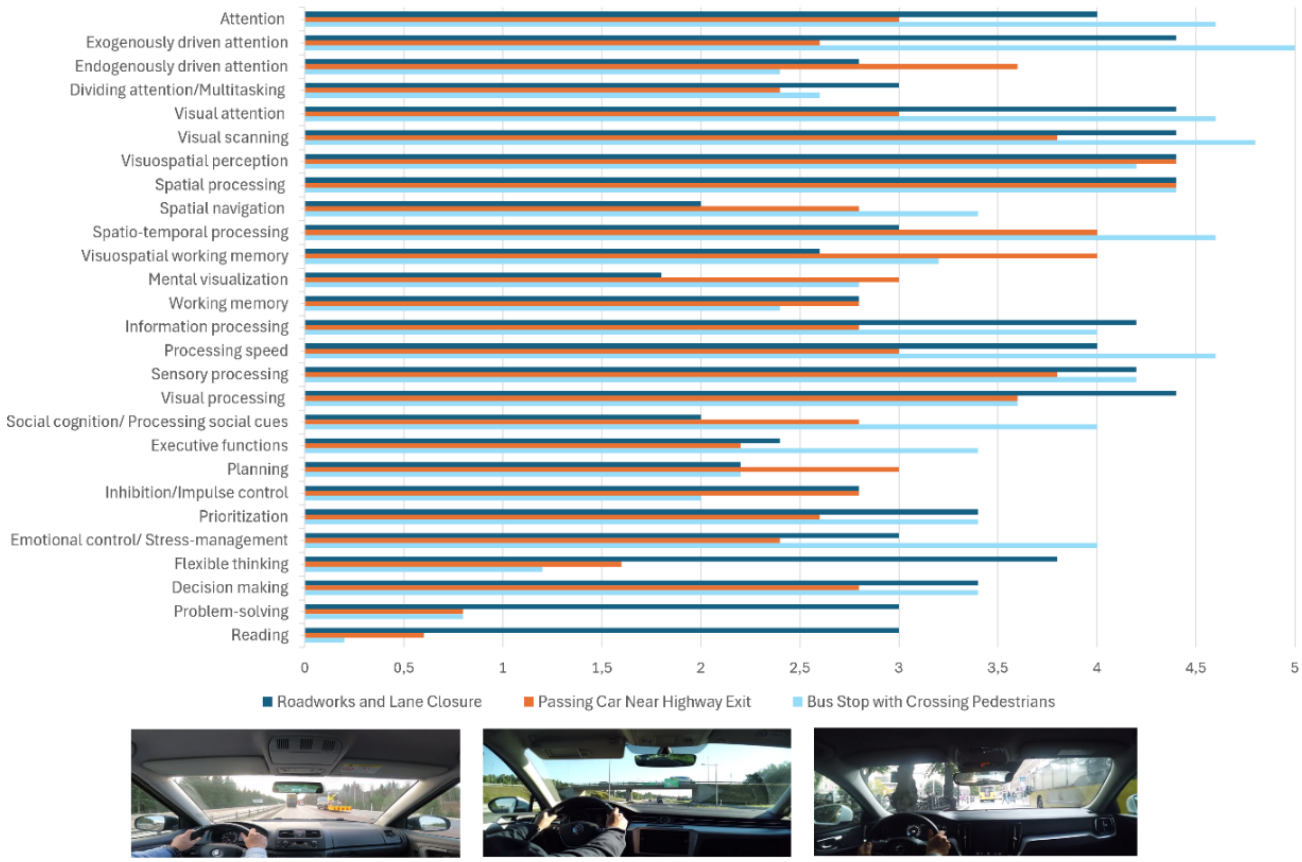


Figure1. Comparing cognitive processes involvement across three videos

Discussion and Conclusion

This study examined the extent to which cognitive processes could be classified in Swedish hazard perception test items by cognitive scientists. The results indicate that various cognitive processes at different levels are required for each test item when using the Barragan et al. (2021) taxonomy of cognitive processes involved in hazard perception in driving. However, even with a list of relevant processes, it proved difficult for the raters to determine which specific processes were involved in each scenario and to what degree. This suggests that, while Barragan et al.'s (2021) framework provides a useful theoretical foundation, applying it to specific hazard perception items may require more structured coding procedures than those used in the present study. The low agreement observed for 17 of the 27 videos suggests that coding the level of cognitive process involvement in hazard perception scenarios is, to some extent, inherently difficult. This may reflect both the overlapping, multi-process nature of real-world driving situations and the subjectivity involved in rating them without structured training. This suggests that expert judgment alone may not be sufficient to reliably characterize the cognitive demands of every item. A further limitation concerns the study's reliance on expert judgment and its limited sample size. All raters were cognitive scientists based in Sweden, so their ratings may reflect context-specific research traditions that do not generalize elsewhere, and the panel's lack of

traffic or driving experts meant it lacked an applied perspective on the scenarios. Raters were also not given structured training prior to the task, despite receiving shared process definitions, which may have led to inconsistent interpretation across raters. These findings should therefore be treated as an initial step rather than a conclusive account, warranting further testing with a larger, more diverse rater panel.

In conclusion, cognitive scientists rated a broad range of cognitive processes in the Swedish hazard perception test material, indicating that these scenarios draw on multiple cognitive functions rather than a single process (RQ1). Involvement ratings varied both within and across videos, and most scenarios showed low overall agreement (RQ2). Future research should build on these initial findings using larger, more diverse, and better-trained raters, ideally incorporating traffic experts and behavioral measures, to investigate the involvement of cognitive processes in hazard perception test items.

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The Impact of Tilting Cabins on Visual Attention, Workload and Human Performance in Forklift Operation

by

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Introduction

Forklift trucks are essential in warehouses and distribution centres, where reach trucks are widely used for material handling in confined environments (Babapour et al., 2012; Arias Portela et al., 2024). Operating reach trucks requires sustained visual attention and frequent twisting of the neck and upper body to monitor loads at height, contributing to musculoskeletal strain and high visual demands (Babapour et al., 2012; Flodin et al., 2018). Dividing visual attention between the environment and in cabin displays increases forklift operators' cognitive workload and reduces situation awareness (Wickens, 2002; Endsley, 1995). To address these challenges, tilting cabin technology was introduced by Toyota Material Handling Europe (Toyota Material Handling Europe, n.d.) to align the operator's line of sight with elevated loads and reduce awkward postures. Tilting cabins are ergonomically beneficial, yet their effects on visual behaviour, cognitive load, and in-cabin system use during load handling are unclear. This study examines these during pallet insertion and extraction. Zagermann (2016) highlights eye-tracking as an objective method for assessing visual attention. In line with this, the present study uses Tobii Pro Glasses 3 to measure and analyse drivers' visual behaviour in a tilting cabin.

Aim and Research Questions

This study investigates how tilting cabin technology influences forklift operators' human performance during pallet insertion and extraction, focusing on visual attention and perceived workload. The purpose is to determine if a tilting cabin enhances human performance by optimizing attention distribution between the load and the screen, thereby promoting a safer and more sustainable operating environment. Visual attention is assessed using eye-tracking, and perceived workload using NASA-TLX (Hart & Steveland, 1988), comparing performance with and without a tilting cabin. Given the limited set of available measures, cognitive workload was not directly measured but indirectly inferred through fixation-based eye tracking metrics related to AOIs. The research questions (RQ) are: *RQ1: How does a tilting cabin affect*

operators' visual attention to the in-cabin screen during load handling?; RQ2: How does a tilting cabin affect operators' perceived workload during load handling?

Method

A controlled within subject design was employed to investigate how tilting cabin technology influences forklift operators' visual attention to the in-cabin screen and perceived workload during standardized load handling tasks. Each participant completed the same predefined route under two conditions, which were counterbalanced: tilting cabin enabled and tilting cabin disabled.

Participants

Licensed forklift operators were recruited through Toyota Material Handling Europe (n = 15). Participants who normally wore prescription glasses were asked to use non-multifocal contact lenses to ensure compatibility with the eye-tracking equipment; those unable to do so were excluded. All participants received written and verbal information about the study, provided informed consent, and were informed of their right to withdraw at any time. All procedures complied with the ethical guidelines of the Swedish Research Council.

Equipment

Visual attention was recorded using Tobii Pro Glasses 3 (Tobii), which captured participants' field of view and gaze data during real world operation. Data were processed in Tobii Pro Lab (Tobii (n.d.-b). The experiment was conducted using a Toyota BT Reflex reach truck equipped with two in-cabin screens and a tilting cabin function to be activated depending on condition (Figure 1).



Figure 1: *Toyota BT Reflex reach truck and Tobii Pro Glasses 3*

Procedure

The study was conducted at Toyota Material Handling Europe's operational facility in Mjölby, Sweden. After consent and equipment setup, participants completed a practice lap to familiarize themselves with the vehicle and environment. They then performed a standardized route

involving pallet insertion and extraction at lift heights of 8 m and 11 m. After completing the first route and the subsequent NASA-TLX questionnaire, participants switched to the alternate cabin condition and repeated the procedure.

Data Analysis

Eye-tracking data were analysed with the in-cabin screen defined as the primary area of interest (AOI). Measures included total fixation duration, number of fixations, and time to first fixation. NASA-TLX provided overall and subscale workload scores. Inspection of the paired difference scores indicated that the normality assumption was not met. Therefore, a non-parametric Wilcoxon signed-rank test was used to compare the tilting and fixed cabin conditions. Statistical analyses were performed in Jamovi (The jamovi project, 2026), with significance set at $p < .05$.

Results

To address RQ1, three eye-tracking metrics were analysed. Regarding the number of fixations, a statistically significant difference was found ($W = 68.5$, $p = .023$, $rrb = .76$), with fewer fixations on the in-cabin screen in the tilting cabin condition ($M = 22.4$, $SD = 19.4$, $Mdn = 21$) compared to the no-tilt condition ($M = 31.9$, $SD = 29.0$, $Mdn = 27$). For total duration of fixations, no significant difference was observed ($W = 61.0$, $p = .092$, $rrb = .56$), though the tilting cabin showed lower mean and median values ($M = 17.4$, $SD = 23.2$, $Mdn = 11.7$) than the no-tilt condition ($M = 24.4$, $SD = 21.9$, $Mdn = 16.7$). Finally, time to first fixation (TTFF) showed no significant difference (W = 31.0, p = .89, $rrb = -0.06$) between the tilting ($M = 2.20$, $SD = 3.22$, $Mdn = 1.13$) and no tilting condition ($M = 3.47$, $SD = 7.10$, $Mdn = 1.57$).

Regarding RQ2, the analysis of NASA-TLX revealed significantly lower total workload when using the tilting cabin ($W = 65.5$, $p = .041$, $rrb = .679$), with lower perceived burden in the tilt condition ($M = 27.38$, $SD = 18.2$, $Mdn = 21.6$) compared to the no-tilt condition ($M = 34.93$, $SD = 16.8$, $Mdn = 37.1$). The NASA-TLX subscales showed that the tilting cabin primarily reduced Mental Demand ($M = 30.42$ (SD = 21.3) vs. $M = 44.16$ ($SD = 28.0$)) and Effort ($M = 25.0$ ($SD = 19.2$ vs. $M = 39.75$ ($SD = 26.3$)). Other dimensions also showed lower mean scores in the tilt condition, however not significant, and Performance were nearly identical ($M = 35.0$ ($SD = 27.4$) vs. $M = 33.33$ ($SD = 24.1$)). No difference in task compliance was observed.

Discussion

The results suggest that tilting cabin technology positively affects both visual behaviour and perceived workload during reach-truck operation. For RQ1, operators made significantly fewer

fixations on the in-cabin screen when using the tilting cabin, indicating a reduced need to consult screen information. This likely reflects improved visual access to the load, allowing operators to rely more on direct environmental cues rather than shifting attention between the load and displays, in line with established models of visual attention and situation awareness (Endsley, 1995; Wickens, 2002). Total fixation duration and time to first fixation were non-significant but lower in the tilt condition, suggesting reduced in-cabin system dependence, not altered initial strategies.

For RQ2, the tilting cabin led to a significantly lower overall perceived workload, driven mainly by lower Mental Demand and Effort. This supports earlier research showing that improved ergonomic design can reduce cognitive and physical strain in forklift and reach truck operation (Babapour et al., 2012; Flodin et al., 2018). The lack of Performance differences suggests the technology eases task execution without affecting perceived success. The lower workload may reflect improved physical ergonomics and more efficient visual attention allocation, as better load visibility reduces the need for visual reorientation. Future studies could use additional eye-tracking metrics, e.g., gaze transitions between load, forks and screen to explore this further.

Conclusions

The results show that a tilting cabin reduces operators' visual reliance on the in-cabin screen, as indicated by fewer fixations during load handling, while not changing when screen information is first consulted. This suggests more efficient visual attention allocation when the operator's line of sight is aligned with the load. In addition, the tilting cabin significantly reduced perceived workload, mainly by lowering Mental Demand and Effort, without affecting perceived performance. Together, these findings indicate that tilting cabin technology supports easier and less cognitively demanding load handling while maintaining task effectiveness.

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Predictive Support for Stopping and Parking: Technologies, Evaluation Methods, and Gaps for Remote Train Driving

by

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Introduction

Stopping and parking manoeuvres are critical phases of vehicle operation, requiring precise control, accurate distance estimation, and continuous perception–action coupling (Takahashi, 2013). In many transportation systems, failures or inefficiencies during these manoeuvres can negatively impact safety, comfort, and operational performance. These challenges are amplified in contexts where sensory information is restricted or delayed, such as in remote or automated operation (Dybvik et al., 2021).

To address these challenges, a variety of predictive and assistive technologies have been proposed, including Head-Up displays (HUDs), augmented reality (AR) overlays, visual guidance lines, and other driver assistance interfaces. While individual studies have investigated the effectiveness of such technologies, the literature remains dispersed across transportation domains, methodological traditions, and evaluation metrics. As a result, there is limited shared understanding of which technologies have been explored, how they have been evaluated, and where critical research gaps remain. The present paper addresses this gap by providing a structured review of predictive technologies designed to support stopping and parking manoeuvres in ground transportation.

Aim and research question

By synthesising both technological approaches and evaluation methods, the review aims to inform future research and design efforts, particularly in remote train driving, where evidence is scarces. This review is guided by the following research questions:

RQ1: What predictive technologies have been used to assist stopping and parking manoeuvres in ground transportation?

*R*Q2: *Which methods have been used to evaluate the effect on parking performance and perception of the technologies?*

Method

The review followed the PRISMA and PRISMA-S guidelines to ensure transparency, reproducibility, and bias control. Searches were conducted in Scopus, the ACM Digital Library, and TRID. The following query was applied across the databases:

" ("train driving" OR "rail teleoperation" OR "remote train driving" OR "train simulator" OR "ATO" OR "car" OR "bus" OR "tram" OR "boat" OR "ship" OR "transportation system") AND

("predictive display" OR "augmented reality" OR "AR" OR "head-up display" OR "HUD" OR "guidance lines" OR "parking aid" OR "driver assistance system" OR "visual aid" OR "assistive interface" OR "ADAS") AND

("user experience" OR "usability" OR "UX" OR "trust" OR "workload" OR "cognitive demand" OR "performance" OR "stopping distance" OR "time to collision" OR "TTC" OR "reaction time" OR "NASA-TLX" OR "UEQ" OR "Stopping Accuracy").

Studies included followed the characteristics defined using the PICO framework: (i) Population: human operators performing stopping or parking maneuvers on ground transportation systems; (ii) Intervention: technological systems supporting parking/stopping maneuvers; (iii) Comparator: use versus non-use of support or predictive technology, comparisons across the same and different systems in different conditions, and comparison between guidance lines and other technologies; (iv) Outcome: objective performance measures and subjective experience evaluations. Apart from these criteria, studies included were peer-reviewed journal or conference papers, in English, published from 2010 onward, conducted in simulator, prototype, or real-world settings, with evaluation involving human–system interaction.

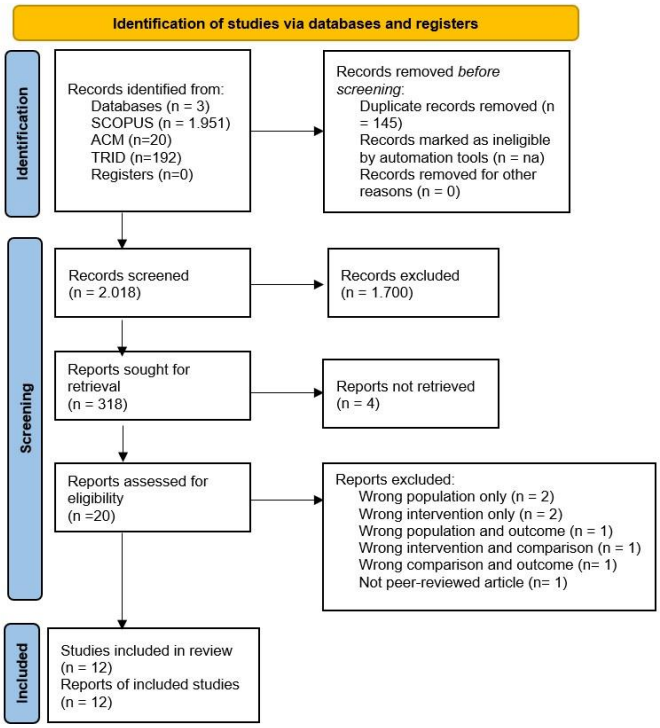


Figure 1. *Prisma Flow Diagram*

Results

Technologies and Transportation Contexts

The reviewed studies predominantly focus on automotive contexts, with head-up displays and augmented reality visualisations being the most common technologies for parking, stopping and following manoeuvres. No study was found to involve train driving, and only one study addressed remote operation.

Evaluation Measures

The most representative objective variables were those related to time (e.g., reaction time, time to park, time to collision), those related to distance (e.g., first braking distance, distance from the obstacle), and those related to speed (e.g., acceleration, velocity). Instead, the most representative subjective variables were those related to user experience (e.g., utility, satisfaction), those related to technology acceptance, and those related to cognitive workload.

Identified Gaps

Notably absent from the literature are studies addressing: (i) Predictive support for stopping or parking in rail transport; (ii) Remote or teleoperated train driving; (iii) Guidance line-based support technologies in this context.

Discussion and Conclusion

The findings highlight a strong concentration of research on automotive applications and visual display technologies, with limited transfer to other transportation domains. The absence of studies in rail and remote driving contexts suggests that existing design and evaluation practices may not directly generalise to these domains. The review further reveals a lack of holistic evaluation, in which the design of predictive technologies, as well as both behavioural and experiential outcomes related to their use are considered.

This paper provides a structured overview of predictive driver support technologies for stopping and parking manoeuvres in ground transportation. By synthesising existing approaches and identifying key gaps, the review offers a foundation for future empirical studies addressing emerging challenges in remote and automated train operation. Based on the review, future studies should: (i) extend predictive support research to rail and remote driving contexts; (ii) systematically compare predictive visual aids, including guidance lines; (iii) adopt mixed methods combining performance-based and experiential measures.

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The Effect of a Tilt Cabin Reach Truck on Muscle Activation and Perceived Driver Experience

by

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Introduction

Forklift trucks are essential in industrial and logistics operations but often involve physically demanding tasks that contribute to musculoskeletal strain, particularly in the neck, shoulders, and back. In Sweden, forklift operators report high rates of work-related musculoskeletal disorders (Flodin et al., 2018). One key ergonomic challenge is repeated upward head tilting during high-lift work. To address this, Toyota Material Handling Europe introduced tilting cabin designs to promote a more neutral working posture. A recent Swedish study found that operators perceived tilting cabins as reducing neck and shoulder strain and improving visibility during high-lift tasks (Turén & Thorslund, 2026). While cabin tilt is primarily intended to reduce physical strain, it may also influence cognitive aspects of task performance. Improved visibility can support visual information gathering, reduce the effort required to monitor loads at height, and alter how operators engage with the task environment. Understanding the relationship between objective physiological measures and operators' subjective experiences is therefore relevant not only from an ergonomic perspective but also for understanding perception and task engagement in applied work settings. Building on this, the present study combines subjective evaluations with objective measures of muscle activation in a controlled setting, where participants performed identical tasks with and without cabin tilt. The objective methodology follows Ahlström et al. (2019), using surface electromyography (sEMG) to assess muscle activation. Earlier work on reach trucks showed that tilt reduced head extension by approximately 20% (Cederqvist, 2000). In contrast, sEMG provides a more direct assessment of muscular load rather than observable movement alone.

Aim and research questions

The aim of this study was to evaluate how a tilting cabin in reach trucks affects operators' physiological workload and subjective experience during high-lift tasks compared to a conventional fixed cabin. Particular attention was given to the relationship between measured

muscle activation and operators' perceptions of visibility, comfort, and work performance. The following research questions were addressed: *(RQ1) How does cabin tilt influence activation of the sternocleidomastoid, neck extensor, and upper trapezius muscles? (RQ2) How do working height and cabin tilt interact to affect muscle activation? (RQ3) How do operators describe visibility, comfort, and strain when using cabin tilt?*

Method

This within-group study was conducted at Toyota Material Handling Europe's facilities using one of their trucks, with and without the tilt-function active. Neck muscle activation and operators' experiences was measured through sEMG and semi-structured interviews.

Participants

Twenty-two experienced reach-truck operators were recruited from Toyota Material Handling Europe and its customers. Participants were aged 21–56 years; 13.6% identified as female, 81.8% as male, and 4.5% did not report gender. On average, they reported 31.8 hours of weekly operation and 13.3 years of certification experience. All participants had prior experience with tilting cabins and reported using this function in their current work.

Equipment

The study used a Toyota BT Reflex reach truck equipped with a tilting cabin, with a maximum reach height of 11 m (Figure 1a). sEMG was recorded using self-adhesive Ag/AgCl Neuroline electrodes. A total of 13 electrodes were connected to a Vitaport 3 amplifier. Two electrodes were placed on six individual muscles across three muscle groups, with an inter-electrode distance of 1 cm.

Procedure

Participants were instructed to retrieve a pallet from a shelf at 8 m height, lower it to the ground, and return it to the same shelf. After placing the forks on the ground, the task was repeated with a pallet at 11 m. These are segments of the total task. Participants then returned to the starting position, where the test leader adjusted the truck settings, before repeating the procedure. Prior to the task, all participants performed a maximum voluntary contraction (MVC) for the studied muscles. Participants were counterbalanced such that half began with the tilt setting activated and half without.

Analysis

For the statistical analysis of the sEMG data, linear mixed-effects models (LMMs) were used to examine differences across tilt settings, heights, muscle types, and participants. The models

were based on mean muscle activation per segment, with tilt setting, height, and muscle type as fixed effects and participant included as a random effect. In addition, a thematic analysis was conducted on the qualitative interview data following the approach by Alvinius et al. (2023).

Results

When using data normalised to MVC max and excluding the neck extensors, due to poor MVC normalization reliability and systematic signal saturation, a LMM revealed two significant effects ($p < 0.05$). Specifically, not using tilt resulted in higher overall muscle activation compared to using tilt ($\beta = 0.64814$, $p = 0.000253$), and the trapezius showed lower activation than the sternocleidomastoid ($\beta = -2.72459$, $p <$ 2e-16). Prior to normalizing the data, both the sternocleidomastoid and trapezius muscles exhibited lower activation than the neck extensors sternocleidomastoid: ($\beta = -25.581$, $p <$ 2e-16; trapezius: $\beta = -10.137$, $p =$ 1.69e-05). The interaction between no-tilt and trapezius activation approached significance ($0.05 < p < 0.10$), indicating a tendency toward higher trapezius activation when tilt was not used ($\beta = 6.452$, $p = 0.0505$).

The difference in muscle activation between work at 8 m and 11 m height was calculated by subtracting activation with tilt from the corresponding activation without tilt for each movement. Using the mean activation for each movement, the results showed that the effect of tilt differed across muscles. In 17 of the 36 total movement conditions, the use of tilt resulted in a larger difference in activation between the two heights.

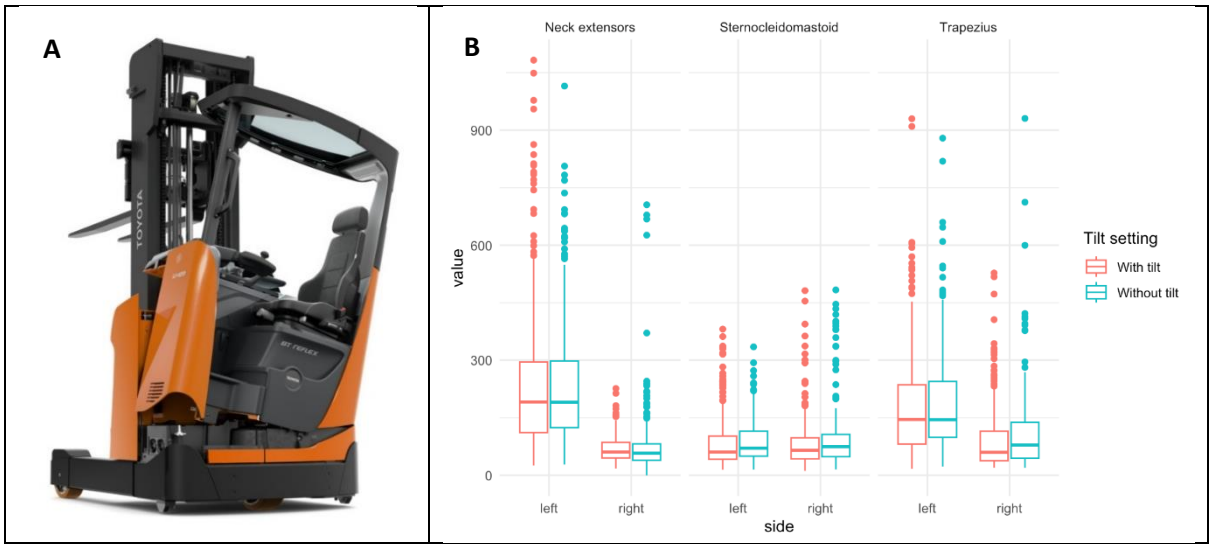


Figure 1: *Toyota BT Reflex (A) and Max value of the 95% percentile of data, not normalized (B)*

Thematic Analysis

Overall, the tilt function was positively received, frequently described as "better" and "more pleasant," with 12 participants expressing a preference for tilt unprompted. In addition, seven participants highlighted improved visibility, particularly noting a clearer view of the sides of

the pallet. The impact on the neck was a recurring theme, often mentioned without prompting. Participants generally perceived the tilt setting as reducing neck strain, and one participant linked the development of tilt cabins to reduced pain over time. Experiences of discomfort were common: five participants spontaneously mentioned neck pain, and when prompted, a total of 11 participants identified the neck as an area of discomfort. The shoulders and back were each mentioned by three participants. Some participants also noted that discomfort tends to occur early in a work period, either at the beginning of a career or when they return after a break and need to readjust. The main drawback of the tilt cabin, mentioned by more than one participant, was that it requires more space when lifting, which may pose challenges in narrow aisles.

Discussion

Although a statistically significant overall reduction in activation was observed with tilt, the effect was not consistent across muscles and heights. While slightly higher activation without tilt is observed in the maximum values. This finding is not consistent with previous work reporting a 20% reduction in neck extension with tilt (Cederqvist, 2000).

The comparison between 8 m and 11 m working heights suggests that tilt may lead to larger differences in activation between heights, rather than uniformly higher or lower activation. This may reflect improved visibility, enabling operators to engage more actively with tasks at greater heights, where perceived risk may be higher.

The thematic findings align with earlier research, as participants expressed a clear preference for the tilt setting and frequently reported improved visibility and reduced neck strain. Interestingly, these perceived benefits were more pronounced than the differences observed in the sEMG measures. This suggests that subjective experience and physiological workload may capture different aspects of operator performance. Improved visibility may enable operators to monitor loads more comfortably, reducing the need for awkward head and neck postures. Consequently, operators may perceive the task as easier and less strenuous despite only modest changes in muscle activation. This may explain why participants consistently reported improved comfort and reduced neck strain with tilt, whereas the sEMG results showed relatively small differences. The findings suggest that ergonomic benefits may arise from improved posture and visual access rather than substantial reductions in muscular workload.

Conclusions

Tilted cabins did not produce a consistent overall reduction in muscle activation, although normalised data suggest a tendency toward lower activation with tilt. Instead of uniform effects, tilt was associated with greater variation in activation between working heights, indicating changes in task engagement rather than load reduction alone. Operators nevertheless strongly preferred tilt, reporting improved visibility and reduced neck strain. The mismatch between subjective and sEMG results suggests that perceived benefits may stem from improved posture and visual behaviour rather than directly lowered muscle activity. More broadly, the findings indicate that workplace design may influence operators' perception of the task and their engagement with visual information, highlighting the value of combining physiological and subjective measures when studying human performance in applied work settings.

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Think-Aloud Commentary and Visual Attention in Young Drivers: A Simulator Study

by

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Introduction

Young drivers are disproportionately involved in crashes (Banz et al., 2019). Although they represent a small share of the population, they account for a significantly more traffic fatalities in the EU. This overrepresentation is commonly attributed to limited driving experience and an overestimation of driving ability (Thorslund et al., 2024). Limited experience reduces exposure to hazardous situations, hindering the development of mental models needed for effective situational awareness (SA) (Horswill & McKenna, 2004). SA comprises perceiving relevant information, understanding its meaning, and projecting future events (Endsley, 1995). Previous studies show that novice drivers display narrower visual scanning patterns than experienced drivers, typically focusing straight ahead and overlooking potential hazards in the periphery (Underwood et al., 2003). Think-aloud (commentary driving) has been proposed to improve novice drivers' SA. Research indicates that commentary can enhance visual scanning and hazard detection (Isler et al., 2009; Crundall et al., 2010). However, most studies rely on video-based tasks that do not capture the cognitive demands of driving. It therefore remains unclear whether these benefits transfer to more realistic driving contexts. This study addresses this gap by using eye tracking to examine the effects of think-aloud commentary in a driving simulator.

Aim And Research Questions

The aim of this study is to investigate whether think-aloud commentary improves young drivers' situational awareness in a realistic driving simulator environment with the following research questions:

RQ1: How does think-aloud affect young drivers' visual attention while driving in a simulator?

RQ2: How does think-aloud influence young drivers' self-awareness and perceived risk during driving?

RQ3: How do young drivers perceive the use of think-aloud while driving?

Method

The study employed an experimental between-groups design to examine whether think-aloud commentary can enhance situational awareness (SA) in young drivers. Participants were assigned to either a think-aloud group or a control group.

Participants

Participants were 24 young drivers (18–30 years; M = 22.9, SD = 2.83) recruited voluntarily from Linköping University. All had held a driving licence for up to seven years. The sample included 11 females and 13 males, with 12 participants assigned to each group. Participants driving frequency was similar across groups, with most driving a few times per year or monthly.

Equipment

The study used a fixed-base driving simulator with Logitech controls, adjustable seating, and a seatbelt (Figure 1a). The simulator featured three curved 32-inch screens providing a 120° field of view, three functional rear-view mirrors, and traffic and engine sounds delivered through headphones. It ran on the Skillster platform, which provides a realistic traffic environment and records driving behaviour and response times. The scenario was set in a Swedish residential urban area, including typical road users and infrastructure elements, with vehicles and pedestrians appearing at identical locations for all participants. Following Wynne et al. (2019), the setup qualifies as medium-fidelity. Eye-tracking glasses (Tobii Glasses X) were used to record glance behaviour. The system uses Pupil Center Corneal Reflection (PCCR), with an accuracy of approximately 1.5-2.5 degrees and a sampling rate of 120 Hz. Eye-tracking data were analysed using Tobii Pro Lab, enabling identification of fixations based on foveal vision.

Procedure

Upon arrival, participants received study information, provided informed consent, and were assigned to either a control or think-aloud group, balanced for driving frequency. All participants completed a simulator warm-up session before the eye tracker was calibrated and recording started. Participants in the think-aloud group were instructed to verbalise their observations while driving. The test scenario lasted approximately 6-7 minutes. Afterwards, participants completed a self-awareness questionnaire adapted from Thorslund et al. (2024) and rated their driving performance and perceived risk. Participants in the think-aloud group also evaluated their experience of using the think-aloud method.

Data Analysis

Participants' visual attention was analysed using heatmaps and fixation measures. Gazes were annotated onto panoramic snapshots of selected time of interest (TOI). Heatmaps were generated to visually compare gaze distributions between the groups (see fig. 1b-e). In addition, the total duration of passing through each TOI and average fixation duration within each area of interest (AOI) were calculated and compared between groups using the Mann-Whitney U test. Responses from the self-awareness questionnaire were analysed using a Wilcoxon signed-rank test, with a significance level set at $p < .05$, to examine pairwise comparisons in self-assessed driving performance and perceived risk. Finally, a thematic analysis was conducted to explore how participants in the think-aloud group experienced verbalising their thoughts while driving.

Results

RQ1: Figure 1 presents heatmaps of each group's gaze data while participants passed through two different TOIs: a zebra crossing and a three-way intersection where a car was approaching from the right. For neither of the TOIs was there a difference between the groups in the time taken to pass through the TOI. Furthermore, there were no differences in average fixation duration between the groups in any of the AOIs. However, the think aloud group showed a tendency toward longer average fixation durations in the left-window AOI during the zebra-crossing TOI (M = 207.5ms, SD = 56.1) compared with the control group (M = 155.6, SD = 64.1). This difference approached statistical significance, $U = 25$, $p = .063$, $r = .50$.

RQ2: A Wilcoxon signed-rank test indicated that participants in the control group perceived their driving skill before the test ($Mdn = 3$, $IQR = 1$) as significantly better than their perceived driving skill during the test ($Mdn = 3.5$, $IQR = 2$), $W = 0$, $p = .031$. Additionally, no differences were found between perceived risk and perceived driving skill before the test or during the test. For the think-aloud group, no differences were found between perceived driving skill before and after. Neither were there any differences between perceived risk and perceived driving skill before or during the test.

RQ3: Participants reported mixed experiences with the think-aloud method. Those who found it difficult described it as unnatural, hard to maintain, and challenging to know what to say, sometimes leading to repetition or reduced verbalization over time. In contrast, participants who reported more positive experiences felt that think-aloud increased their awareness and occasionally helped them stay focused. Some also noted that decision-making felt largely

automatic. One participant found it very easy and was already familiar with using think-aloud while driving.

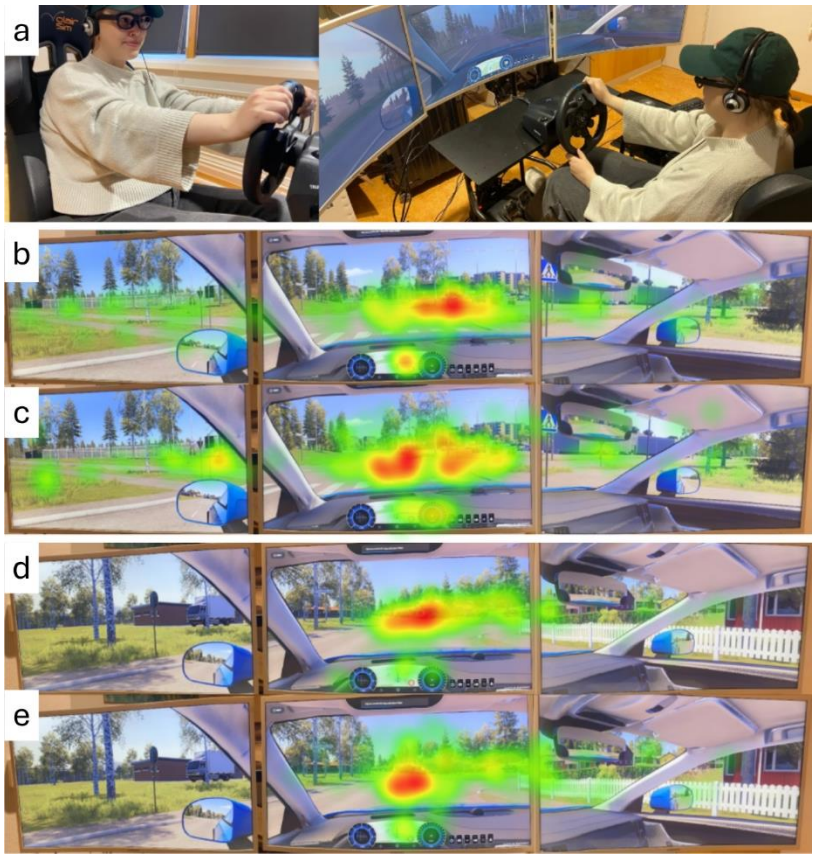


Figure 1: Setup with driving simulator and glasses (a). Heatmap of gaze during the zebra crossing TOI for the control group (b) and think -aloud group (c). Heatmap of gaze during the priority-to-the-right TOI for the control group (d) and think-aloud group (e).

Discussion and conclusion

This study investigated whether think-aloud could enhance young drivers' situation awareness (SA) in a driving simulator. The results revealed no significant differences between the think-aloud and control groups in visual attention. However, the think-aloud group showed a tendency toward longer fixations in the left-window AOI at the zebra-crossing TOI, suggesting that verbalisation may encourage broader visual scanning, which is consistent with earlier studies (Isler et al., 2009; Crundall et al., 2010). Furthermore, only the control group showed a decrease in perceived driving performance after the test, a finding that aligns with previous research suggesting that young drivers may overestimate their driving ability (Thorslund et al., 2024). Participants generally perceived think-aloud positively and reported that it increased their awareness of the driving situation, although several also described difficulties maintaining

a consistent verbalisation process throughout the drive. Future research with larger samples and additional eye-tracking measures, such as time to first fixation and saccade behaviour, is needed to further investigate the potential effects of think-aloud on situation awareness. It may also be valuable to examine the cognitive load associated with maintaining concurrent verbalisation during driving. Although the present findings do not provide conclusive evidence that think-aloud improves visual attention, they suggest that the method may support self-monitoring and situation awareness without adversely affecting driving performance. Therefore, think-aloud appears to be a promising and cost-effective tool for supporting young drivers' situation awareness in simulator-based training.

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Head-Up Displays for ETCS

by

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Introduction

Sweden's railway system is a critical component of national transport, carrying large volumes of passengers and freight daily. To maintain efficiency and safety, Sweden is transitioning from the ageing Automatic Train Control (ATC) system to the European Rail Traffic Management System (ERTMS), with ETCS as its signalling subsystem. While ETCS enables increased capacity and cross border interoperability, it also alters the train driver's work environment by moving essential information from trackside signals to an in-cab driver–machine interface (DMI). Recent simulator studies indicate that ETCS may reduce drivers' outward visual attention, leading to increased in-cab glances, particularly at stations and level crossings (Olsson et al., 2025). This is concerning given that most railway accident fatalities involve unauthorised persons on the track or level crossings (Eurostat, 2025). Reduced outward attention may therefore compromise hazard detection in high-risk areas. One potential approach to mitigate this effect is the use of Head Up Displays (HUDs), which present critical information within the operator's forward field of view. HUDs have been shown in other transport domains to support attention allocation by allowing simultaneous access to external and system information (Ward & Parkes, 1994; Newman, 1995). However, no prior studies have designed and empirically evaluated an ETCS Level 2 HUD using eye tracking. This study investigates whether integrating a HUD with ETCS can improve train drivers' visual attention in safety critical environments. A HUD may reduce attentional switching between external signals and in-cab displays, thereby supporting visual attention allocation.

Aim and Research Questions

The aim of this study is to design and evaluate an ETCS Level 2 Head-Up Display for train drivers and to examine its effects on visual attention from a safety perspective using a driving simulator and eye-tracking. For this, the following research questions (RQ) were formulated: *RQ1: How does an ETCS-HUD affect train drivers' visual attention?; RQ2: What information*

should an ETCS-HUD present to support safe driving?; RQ3: How should ETCS-HUD visualisation and placement differ from the standard ETCS?

Method

The study followed a two-phase design. First, an exploratory design study was conducted to develop an ETCS Head Up Display (HUD). Second, the final HUD prototype was evaluated in a simulator pilot study using eye tracking. Visual attention results were compared with previously collected ETCS non-HUD data from Olsson et al. (2025), forming a between subjects' experimental comparison.

Participants

Participants were recruited through convenience sampling and were required to be current, former, or trainee train drivers with knowledge of ETCS Level 2. Participation was voluntary and compensated with ice cream vouchers.

Phase 1: HUD Design

Three experienced train drivers with a mean age of 41.6 years and train driving experience of 17.6 years participated in semi-structured interviews and a design workshop. All had experience with ETCS Level 2, including one with simulator-only exposure. Sessions were conducted online via Microsoft Teams. A Research through Design approach was used, with iterative prototyping in Figma informed by literature, usability heuristics, and expert feedback. Interviews focused on HUD content, information density, visualisation, colour usage, and placement. Based on interview findings, multiple design proposals were refined and discussed in a group workshop, where participants collaboratively selected and modified a preferred design.

Phase 2: Simulator Pilot

The simulator pilot was conducted in Nässjö, Sweden, using a transportable Regina locomotive simulator provided by VTI, identical to that used by Olsson et al. (2025). A total of 6 train drivers participated, and the mean age was 44 years. All participants had knowledge of ETCS Level 2, and 5 had prior operational experience. One participant was excluded due to data loss. The final HUD prototype was rendered directly on the simulator display, with transparency and colour choices used to approximate optical HUD characteristics. Eye movements were recorded using head-mounted eye-tracking glasses from Pupil Labs, capturing scene video and gaze coordinates. Visual attention was assessed by analysing the distribution of in-cab versus outward glances, with particular focus on safety-critical scenarios such as stations and level crossings.

Data Analysis

Interview transcripts were analysed using manifest content analysis (Bengtsson, 2016) focused on HUD placement, essential information content, and perceived effects on attention. Based on the interview results, the prototype was refined and subsequently reviewed in a design workshop. Remaining design issues were addressed collaboratively, resulting in a final HUD design used for simulator testing. Eye-tracking data from the simulator pilot were analysed using predefined areas of interest (AOI) covering the HUD, the conventional DMI, and the external forward view. The analysis focused on the distribution of in-cab versus outward glances, particularly in safety-critical situations, to assess whether the HUD improved visual attention.

Results

The final ETCS head-up display (HUD) design for five driving scenarios is presented in Figure 1a-e. Across all scenarios, the HUD was positioned centrally at the bottom of the windshield, with size adjusted to viewing distance. The design includes core ETCS DMI information while adapting layout, colour coding, and information prioritisation to support direct forward viewing.

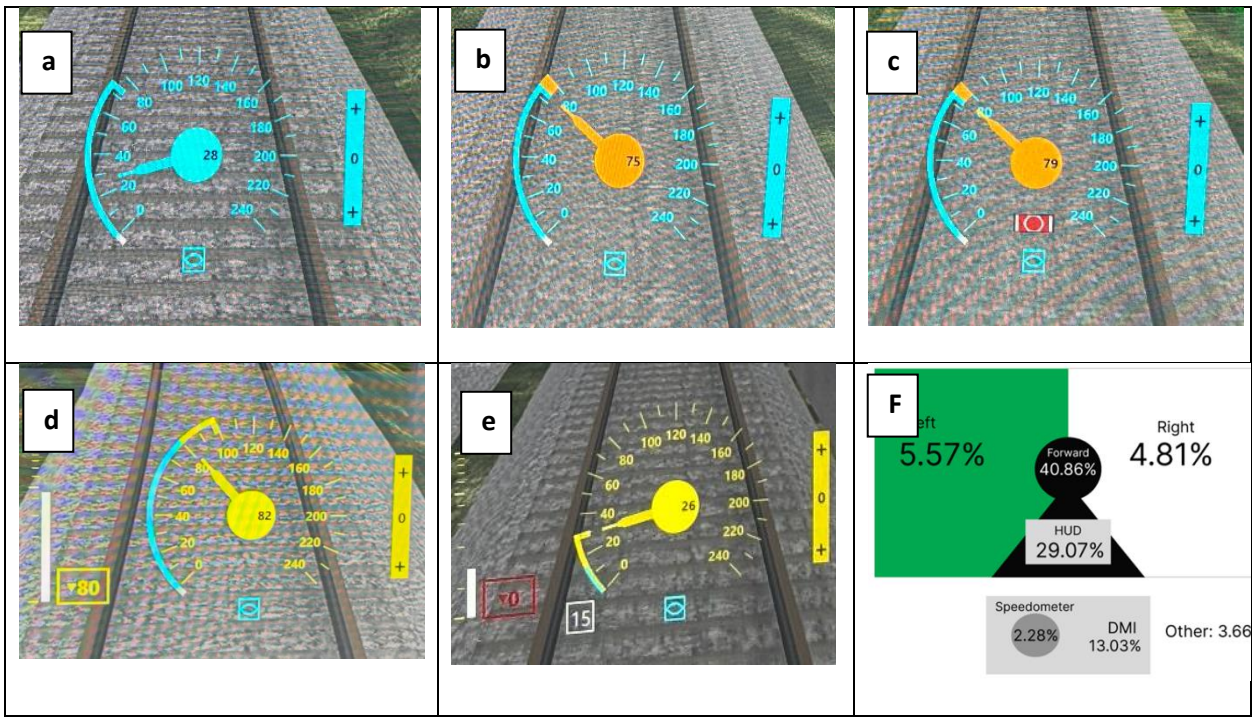


Figure 1: Normal operation (a); Exceeding speed limit (b); Substantially exceeding speed limit – safety brakes (c); approaching a speed decrease (d); Approaching a full stop and release speed (e); Percentages at each AOI with HUD defined as its own AOI (f).

Participants appreciated the clear presentation of speed and current gradient, the placement of elements, and the colour-coded mode changes. Most reported reduced need to consult the in-cab DMI when adjusting speed ahead of restrictions. Preferences for additional information varied, but all except one had a strong willingness to use an ETCS HUD in real-world operations.

Eye-tracking analyses examined glance distribution across AOIs. When the HUD was treated as a separate AOI, no significant difference in total forward glance percentage was found between ETCS and ETCS-HUD conditions (Figure 1f). However, when HUD glances were included in forward glances, the ETCS-HUD condition showed a significantly higher proportion of forward glances compared to ETCS without a HUD ($p < .001$, large effect).

No significant difference was observed in glances toward the DMI excluding the speedometer. In contrast, speedometer glances were significantly reduced in the ETCS-HUD condition compared to ETCS alone ($p < .001$), indicating that speed monitoring was effectively transferred to the HUD. Due to the small sample size, the findings should be interpreted with caution.

Discussion and Conclusions

This pilot study explored whether an ETCS-HUD could support attention to the forward view while providing essential system information. Eye-tracking results showed that when HUD glances were considered part of forward viewing, the ETCS-HUD condition produced a significantly higher share of forward-directed attention compared to ETCS alone. When the HUD was treated as a separate area of interest, no significant reduction in forward attention was observed, suggesting that the HUD redistributed attention away from less relevant areas rather than diverting it from the track ahead. Although HUDs may risk cognitive capture (Gish & Staplin, 1995), the forward view still received a larger proportion of visual attention than the HUD, indicating limited attentional cost.

Survey responses further supported these findings. Participants generally appreciated the HUD design and reported reduced reliance on the in-cab DMI, particularly for speed monitoring, which was reflected in fewer speedometer glances toward the DMI. At the same time, several participants expressed a desire for richer planning information, especially regarding gradients, highlighting a balance between simplicity and informational completeness.

Most participants preferred the tested HUD placement slightly below the natural forward line of sight, consistent with automotive HUD research (Topliss et al., 2019). Overall, the results

suggest that the proposed ETCS-HUD can reduce DMI dependence and support forward-oriented attention. However, altered gaze behaviour cannot be equated with improved safety. As a pilot study with a small sample, further research is needed to assess effects on driving performance, safety, and long-term use under operational conditions. Given the small sample and exploratory design, the findings should be interpreted with caution.

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Abstracts

Interactive game for learning essential driving knowledge

by

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Since the year of 2000, the overall pass rates for both the theoretical and practical driving tests in Sweden have declined significantly. It remains unclear how individuals with neurodevelopmental disorders (NDDs) perform on these tests, however, many NDDs, such as ADHD and dyslexia, are associated with learning difficulties that may affect the driver education process. Previous research indicates that drivers with NDDs often find the theoretical part of the licensing process particularly challenging.

The purpose of this study was to develop an interactive game that supports theoretical learning during the driver licensing process, with a particular focus on NDD-friendly design. In this study, NDD-friendly design refers to adapting the learning experience to reduce unnecessary cognitive demands and make theoretical information clearly accessible through visual, interactive, and context-based presentation. Rather than relying primarily on written text, the game allows users to practice theoretical knowledge within simulated traffic situations.

The game features videos of simulated traffic scenarios combined with multiple-choice questions. As each video plays, it pauses at specific situations and presents a question about what the driver should do or consider. This design connects theoretical concepts to concrete traffic situations and allows users to actively apply their knowledge rather than only reading or memorizing theoretical information.

The game was evaluated through qualitative user testing with 20 university students recruited through convenience sampling. Participants first explored the game freely for five minutes and then took part in a semi-structured interview. The evaluation focused on usability and perceived usefulness, including participants' perceptions of the overall design, consistency, missing features, and the game's potential to support learning of theoretical driving knowledge. Background questions concerning previous experience with driver theory and study methods were also included to contextualize participants' responses. Interviews were audio-recorded with consent, transcribed, and used as qualitative data for the evaluation.

Results from the user testing indicated that the game was experienced as user-friendly and perceived as a helpful alternative for practicing theoretical driving knowledge. Future research should further develop the NDD-friendly design and evaluate the game with larger and more diverse samples, particularly individuals with NDDs, since this was not possible in this study. Future studies could also investigate how the game affects learning outcomes across different cognitive profiles. Overall, the game provides an interactive and visual approach to practicing theoretical knowledge essential for driving and illustrates how NDD-friendly design can be incorporated into digital learning tools for driver education.

How Does Spatial Movement Organise Participation in Toddlers Pretend Play?

by

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Around the second year of life, children begin to engage in pretend play, an activity in which imaginative or make-belief realities are constructed through the shared coordination of meanings (Jackowitz & Watson, 1980). Traditional developmental approaches assume that children's participation in social play emerges alongside the maturation of cognitive abilities (Tetzchner, 2022). However, such perspectives may overlook early forms of pretend play occurring prior to the development of advanced verbal communication. Recent studies suggested that children's pretend play may emerge gradually from earlier embodied social interaction (Fantasia & Nomikou, 2022), and that children's non-verbal behaviours during play interaction were sequentially-organised (Fantasia et al., 2024). In particular, children's spatial configurations seem to be oriented towards seeking engagement and cooperation (Nasi & Karlsson, 2026).

Drawing on these findings, this study explores how spatial movements precede multi-party play episodes, with particular attention to spatial proximity and movement between play facilities. We video-recorded a longitudinal set of 12 ecological play sessions between 10 toddlers (aged 19–28 months) attending Kindergarten. The broaden aim of the study was to identify how children multimodally coordinate spatial orientations to initiate play instances seeking co-presence with another child. Although data analyses are currently ongoing using

ELAN Version 7.1 (ELAN, 2026), a preliminary exploratory analysis using 50% of the recordings was performed on small groups of children playing in the same room including 4 to 6 distinct play areas. Analyses showed that 39.5% of observed approaches to a play area occurred when at least one other child was already present, and 6% occurred when two or three children were already present. These preliminary findings provide an initial characterization of how children move between play areas and form groups. Further analyses of the full dataset will examine children's co-location, movements, posture, and spatial orientation, to provide a more comprehensive picture of their interpersonal coordination. Findings from this research may provide insight into how young children's group participation may emerge before cognitive maturation through embodied and spatial coordination.

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Workslop Erodes Communal Work Engagement

by

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The current hype towards artificial intelligence (AI) is often framed through efficiency gains; faster completion, higher outputs, and reduced labor costs. This framing encourages workslop; using AI for creating surface-level acceptable work without assuring quality (Lim & Jun 2026). Rather than eliminating work, these practices redistribute, obscure and amplify it. This can negatively affect work engagement (WE), a positive state of experience at work, characterised by dedication, vigor, and absorption (Schaufeli et al., 2002). WE is a cornerstone of sustainable work (cf. Gürbüz et al., 2023), requiring cognitive and emotional resources to meet job demands (Bakker & Demerouti 2007) and a perception of meaningful work (Kahn 1990). Studies show that use of AI at work can both support and limit WE in nuanced ways (cf. Liang et al., 2022, Li et al., 2024, Bergqvist et al., 2026). In this position paper, we argue that workslop passes on non-meaningful cognitive processing across teams and organisations and undermines communal WE.

Workslop is hastily produced AI-mediated work that superficially meet requirements but erode value by lacking internal rigour and external grounding (Kommers et al., 2026, Lim & Jun 2026). It passes through organisational workflows that lack thorough validation of quality. Aligning with the ironies of AI (Endsley, 2023), as outputs look sufficient at first glance, they require time, competence, and cognitive effort to interpret, validate, and revise (Lim & Jun 2026). The effort of cognitively managing such flaws of technology like AI is often invisible and unrecognized in mundane work (Fox et al., 2023, Justesen & Plesner, 2024).

Many workers anticipate AI-driven efficiency to free up time for meaningful tasks (Bergqvist et al., 2026). While some individual workers have become more productive through AI use (cf. Noy & Zhang, 2023, Brynjolfsson & Raymond, 2025), others are becoming ineffective (Simkute et al., 2025). Further, workers often experience a higher density of cognitively demanding work due to increased communication and administration (ActivTrak 2026), fewer simple tasks, and raised organizational expectations (Bergqvist et al., 2026). Additionally, workslop accumulates as work debt that compounds over time (cf. Storey, 2026, Sawada et al., 2026). AI-driven efficiency thus resembles Jevon's paradox: AI-driven efficiency raises the volume of work and demands instead of reducing them (Fiehler 2026).

Further, workslop shifts labor and its consequences onto others (Madsen & Puyt, 2025, Kommers et al., 2026). For example, in research practice, the burden of manuscript workslop falls on reviewers (cf. Helgeson et al., 2025, Kashou et al., 2025) and in teaching the focus becomes detecting students' use of AI instead of giving meaningful feedback (Corbin et al., 2025). The resources of experts are then used on undervalued and less meaningful but cognitively demanding tasks instead of high quality meaningful work, which can negatively impact WE (Bakker & Demerouti, 2007). While the creation of workslop might not affect the workers that created it directly, its negative consequences on WE are then displaced to coworkers that need to compensate for it.

We thus argue that workslop, through forwarding increasing cognitive demands and perceived insignificance, obstructs sustainable work and erodes communal WE.

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How Do We Study Consciousness? Issues in NCC and Unconscious Processing Research

by

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In cognitive neuroscience, two central goals are to identify the neural correlates of consciousness (NCCs) and to determine the extent of unconscious cognitive processes. Both topics remain actively debated. A common approach to studying NCCs involves presenting stimuli so that participants are aware of them on some trials and unaware on others. Neural differences between these conditions are interpreted as NCCs. For example, visual awareness negativity (VAN), measured with EEG, has been proposed as an NCC. However, neural differences between aware and unaware trials may reflect processes other than consciousness, such as attention. In addition, because awareness is typically measured via self-report, it is difficult to rule out response biases. Another approach focuses on demonstrating unconscious cognitive processes, such as unconscious working memory. Here, analyses are restricted to trials in which participants report no awareness, and any observed neural responses are interpreted as unconscious. For example, research suggests that the contralateral delay activity

(CDA), an EEG-index of visual working memory, can be elicited by stimuli reported as unseen. However, this approach also faces challenges, including methodological issues such as regression to the mean due to trial selection. A recent proposal is to study functional dissociations. This approach is convincing if results represent qualitative differences, but such differences have been found only in limited situations. I will review these continuing and new challenges in consciousness research and illustrate them with examples from EEG studies conducted in my lab.

Interaction with roles rather than agents

by

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In research investigating human interaction with non-human (and in particular artificial) agents, much attention has been paid to what kind of agent the human is interacting with and to what extent (or in what way) that agent is human-like (e.g., Lagerstedt and Thill, 2020). However, although this strategy can often be quite useful and informative, it also overgeneralises an overly simplified view on human-human interaction. The way humans interact with other humans depend largely on what role (in the sense of Goffman, 1959) the other human is performing at that particular instance, as well as the context in which the interaction happens, and the roles that are performed can change dynamically throughout an interaction. This phenomenon is particularly under-explored in many discussions related to human interaction with social robots (Healey et al., 2023). This is not in opposition to, for instance, theories related to the intentional stance in which artificial agents are interpreted *as if* they have more complex human properties (Dennett, 1989), whether or not they are believed to *actually* possess such properties (Ziemke et al., 2015). Our claim is rather that the discussion should more often be at the level of the situated role of the agent rather than universal properties essential to the kind of agent.

For example, in a study where humans were interacting with a virtual assistant (Alexa) in domestic situations (Vanzan et al., 2025), there were several instances when humans were speaking to the Alexa and, mid interaction, made remarks about the Alexa to each other as if it was not there. In the particular situation the humans behaved as if the Alexa was participating

in the interaction, they then seamlessly switched to a kind of back stage interaction where the Alexa's presence was not acknowledged, and they finally switched back to acknowledging the Alexa. We call this "the Butler Effect" to emphasise how such otherwise rude behaviour would not be unreasonable under the right circumstances of human-human interaction. For instance, when dinner guests interact with serving staff, the presence of the staff might only be acknowledged when their roles are relevant for the guests. Framing the phenomenon in terms of interactions between roles should help reduce the excessive exotification of non-human agents, and better access the underlying psychological and cognitive dynamics at hand, while facilitating interpretations of affordances. This re-framing also makes it possible to explore questions related to the degree at which different agents are able to engage with such roles. For instance, the theoretical underpinnings of generative AI might make such interactions difficult, in ways which are unproblematic for social non-human animals.

It is worth mentioning that the role of artificial agents is often highlighted as an important factor when interacting with humans (e.g., Gillet et al., 2024). However, the functional and social roles are often conflated beyond the entanglement of what is found in human-human interaction. By emphasising the butler effect, this entanglement is placed more central in the analysis, making the diversity in kinds of simultaneous roles more salient.

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Intuitive Cognitive Expertise in Child Protection Workers: Environmental Prerequisites for Expertise Development

by

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As this study is still in its early stages, no data are available at this time. This text focuses on evaluating the underlying rationale and strengthening the methodological design.

Background

Intuitive expertise — the ability to make rapid, accurate judgments through automatic pattern recognition — develops reliably in some professions but not others. Firefighters demonstrate robust intuitive capabilities, while clinical psychologists show limited expertise despite extensive experience. This variation likely reflects fundamental differences in learning environments rather than practitioner characteristics. Hogarth's (2001) framework distinguishes ‘kind’ learning environments, which provide outcome regularity, adequate practice opportunities, and valid timely feedback, from ‘wicked’ environments characterized by delayed, ambiguous feedback and obscured cue-outcome relationships. Kind environments enable genuine expertise development; wicked environments foster confident but inaccurate intuitions (Kahneman & Klein, 2009).

Research Problem

Child protection work involves high-stakes assessments of child safety and family functioning conducted under time pressure with incomplete information. Workers must rapidly integrate multiple information sources—parental behavior, child presentation, home conditions, historical patterns—to make consequential decisions about child removal, service provision, and case prioritization. Existing research has explored intuition, sensemaking, and experience-based judgment in child protection practice, but the question of whether child protection work environments support the development of valid intuitive expertise remains insufficiently

examined. Understanding environmental characteristics has direct implications for professional training design, organizational feedback systems, and decision support requirements.

Research Objective

This study systematically characterizes the child protection learning environment to determine its position on the kind-wicked continuum and identify specific features that facilitate or impede expertise development.

Methods

We will conduct structured interviews with child protection workers, and supervisors, systematically assessing environmental dimensions identified in expertise literature. We will include some of the following: *Outcome Regularity*: for example strength of relationships between observable cues and child safety outcomes; *Feedback Validity*: accuracy and completeness of outcome information workers receive; *Feedback Timeliness:* typical delay between decisions and outcome knowledge; *Practice Opportunities*: opportunities to observe case outcomes; repetition of similar assessment situations; *Outcome Base Rates:* frequency of serious adverse events.

Interview data will be qualitatively coded and synthesized to characterize the overall learning environment, identify specific facilitators and barriers to expertise development.

Significance

This environmental analysis will provide the essential foundation for understanding whether and how intuitive expertise can develop in child protection work, directly informing investments in experience-based training versus structured decision tools and environmental redesign to support expertise development

Neuro-steered Hearing: Auditory Attention Decoding Across Gaze Conditions

by

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Auditory attention decoding (AAD) uses EEG to identify which sound a listener attends to, with potential applications in neuro-steered hearing devices. However, neural signals associated with auditory attention may also reflect visual attention, especially when listeners look toward a sound source, in a process called overt attention. Conversely, when the focus of attention is outside visual focus, it is called covert attention. This study focuses on the N2ac, an event-related potential linked to spatial auditory attention. I investigate whether the N2ac tracks auditory attention across covert and overt attention and whether gaze toward a visually congruent sound source relates to sound localization performance. Participants listen to competing human voices pronouncing random numbers in English from laterally separated stereo loudspeakers with the task to report the location of a target number. In designated blocks, listeners are instructed to focus their gaze toward the perceived location of the target (overt attention), and in other blocks are instructed to fix their gaze in a central location (covert attention), using eye tracking to measure gaze. 64-channel EEG will be analyzed for N2ac responses, and common spatial pattern classification on the EEG-data will test whether attended auditory targets can be decoded in the different conditions while visual attention is monitored. I expect N2ac responses during auditory-attention trials and predict that gaze toward a congruent visual source will be associated with more accurate sound localization and higher decoding accuracy. The findings will clarify how visual attention relates to neural measures used in AAD, specifically if EEG-data related to visual attention drives the decoding.

The Scent of a Memory

by

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Odor-evoked autobiographical memories are often perceived as being particularly emotional, but less is known about which perceptual properties of odors shape and influence this emotional strength. The present study investigated whether there exists a relationship between odor-elicited arousal and the perceived strength of autobiographical memories, if arousal alone can predict the emotional strength in these memories and which perceptual odor property that predicts if a memory is recalled at all. Odor-elicited arousal was predicted to both covary positively with and significantly predict perceived strength in autobiographical memories while odor familiarity was hypothesised to significantly predict memory recall.

The study adopted a within-subject design based on repeated measures. In total, 17 men and 16 women participated (N = 33) between the ages of 18-64 (M = 37.91 years, SD = 15.29). The participants were screened prior to the testing to ensure that no olfactory deviations were present at the time, such as any respiratory illness or anosmia. The testing consisted of two tasks: 1) memory task and 2) odor task. Both tasks included exposure to 12 distinct odors presented in Sniffin'Sticks. The selection of odor stimuli was based on a 2x2-axis model constituted by the perceptual properties of pleasantness and edibility, which resulted in the use of e.g. raspberry (pleasant/edible), garlic (unpleasant/edible), clean cotton (pleasant/inedible) and valeric acid (unpleasant/inedible). During the memory task, participants were exposed to the complete odor battery, one odor at a time, and then asked if the target odor had recalled any particular memory. If yes, the participants were asked to describe and rate this memory according to its *vividness, emotional strength, specificity* and *valence.* If no memory was recalled, the next odor was presented. During the odor task, the participants were exposed to the same 12 odors again and then asked to rate them according to *pleasantness, edibility, intensity, familiarity* and *arousal*.

The data was then analysed using Pearson's correlation, multiple regression and logistic regression analysis. Contrary to the hypothesised relationships, the analyses showed no relationship between odor-elicited arousal and emotional strength in autobiographical memories, neither correlational ($r = -.087$, $p = .169$) nor predictive ($F_{5,246} = 0.603$, $p = .698$, $R^2 =$

.012, Adjusted R^2= -.008). Odor familiarity did, however, emerge as the only significant predictor for memory recall (χ^2 (5) = 46.14, p < .001, OR = 1.253, p < .001) and an explorative post hoc analysis found odor intensity to partially mediate this relationship. The mediation analysis was conducted using Preacher & Hayes (2004) procedure with 1000 bootstraps. Further explorative analyses also found odor-elicited arousal and odor pleasantness to possibly overlap which might have obscured the true effect of odor-elicited arousal on autobiographical memories. Odor-elicited arousal was further negatively correlated to the valence of memories. Conclusively, the present study found that odor familiarity increases the likelihood of a memory being recalled, this relationship is also possibly mediated by odor intensity. The role of odor-elicited arousal therefore remains uncertain and warrants further investigation.

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Vocabulary learning as its own reward

by

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What motivates humans to think and act once basic needs are met? Food is easily available and once we eat, we satiate. But the thirst for knowledge only seems to increase as we learn. A body of behavioral and neuroscience studies on curiosity suggest that learning itself may be rewarding and that curiosity reflects anticipation for a learning reward. Earlier findings show that primate brain activity preceding anticipated information is similar to anticipatory activity preceding extrinsic rewards such as juice drops or lottery outcomes. But the core assumption still needs to be tested - whether the brain treats information itself as a reward *pay out*. We tested whether knowledge acquisition in terms of vocabulary learning elicits reward-related brain activity. If so, curiosity can be interpreted literally as a reward-dependent drive for knowledge acquisition with the potential to perpetually motivate human information seeking behavior and learning.

We tested 50 participants in a vocabulary learning task during fMRI, while they rated their response confidence and curiosity about the correct answer before receiving feedback. Participants then rated the learning value of the feedback.
Extrinsic reward circuitry was independently identified using the Monetary Incentive Delay (MID) task. The MID task indicated bilateral ventral striatal (VS) activity (roughly corresponding to the head of caudate) was sensitive to both extrinsic reward anticipation and reward outcome.

In line with the reward account of learning, we found that participant learning value ratings were indicative of subsequent likelihood of vocabulary learning. Because knowledge needs to be incomplete to warrant updating, we expected higher post-feedback activity in the VS following incorrect than correct responses. Consistently, we found that reward-sensitive VS differentiated between correct and incorrect responses both before and after feedback. Specifically, higher VS activity for incorrect than correct responses after feedback suggest learning is tied to reward surprise. Thus, learning may indeed be rewarding, and curiosity might thus reflect anticipation for a learning reward.

Measures of Complexity in Visual Working Memory: Symmetry as Computational Redundancy

by

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Previous research shows that regularity can support compression in visual working memory and that symmetry can improve visuospatial memory (Brady et al., 2009; He et al., 2022). What remains less clear is how different geometric relations within a display can be translated into measures of redundant information and related to the distribution of recall errors. We formalize symmetry in circular arrays of oriented ellipses with algorithms detecting global and local paired-axis relations, rotation, and reflection. For each relation, we quantify its degree and calculate effective length: the number of independent orientation values needed to reconstruct a configuration under the detected constraints.
Ten adults (three women; mean age 29.8 years) completed a delayed-estimation task over three sessions of 600 trials each. Five viewed each stimulus for 200 ms and five for 1,200 ms; a

1,500-ms retention interval preceded continuous orientation report. Symmetry measures were calculated post hoc from the reconstructed stimuli. For each participant, we compared Shannon entropy of binned absolute recall errors across symmetry and effective-length levels with a participant-specific independent-encoding baseline, using permutation tests to evaluate differences.

On average, greater symmetry and shorter effective length were associated with lower recall-error entropy; similar trends appeared in both presentation-duration groups. Participant-level tests identified several low-length conditions with lower entropy than baseline. The descriptive pattern differed across symmetry types: entropy for mirror relations tracked the number of valid reflection axes more closely than effective length, whereas entropy for global and local paired-axis relations tracked effective length more closely.

The contribution is to turn multiple symmetry relations into explicit, stimulus-derived constraints on the number of independent elements and relate those constraints to response uncertainty, beyond showing a general symmetry advantage. These associations are consistent with a compression account of visual working memory; they do not establish that observers used the proposed code. The ten-person sample and correlated structural measures limit broader and type-specific inferences.

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Robot Error and Accountability: Perception of Failure and Expectations in Healthcare Encounters

by

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Although robots are often promoted as time-saving and efficiency-enhancing technologies across various settings, their proneness to error may create challenges in the contexts into which they are introduced and may place additional burdens on human workers (Spektor et al. 2021; Mutlu & Forlizzi 2008). Especially in sensitive environments such as healthcare, robot errors may elicit strong reactions and shape expectations toward other actors or bystanders in the situation (Yu et al. 2024; Nielsen et al. 2023) .

In this pilot study, we explore the effects of an error made by a robotic receptionist in a Swedish vårdcentral (healthcare centre), focusing on how the failure is perceived, how observers emotionally respond to it, and what expectations are placed on a co-present human nurse, the patient, and the healthcare institution.

To investigate this, we created four video scenarios depicting the same situation: a patient enters a vårdcentral and is greeted by a nurse, who directs him to check in with the receptionist. The receptionist provides the patient with a room number and directions, but the information is incorrect. As a result, the patient waits for half an hour, misses the appointment, and must book a new one. The videos manipulate the type of receptionist (robot vs. human) and the gender of the nurse (male vs. female) to explore potential stereotype effects. Each participant is shown two videos, featuring both receptionist types (human and robot) and one of the two nurse gender conditions.

For the pilot study presented here, we conducted semi-structured interviews with seven participants (age=21-58, gender: M=4 F=3) all viewing the female-nurse condition videos. The interview data was then transcribed and further analysed using thematic analysis. The two central themes we identified were mistakes and feelings. Participants tended to interpret the error as a system-related issue when the receptionist was a robot, and a human error that was attributed to the receptionist as an individual when the receptionist was a human. All the participants identified with and expressed empathy toward the patient, while none did so with the receptionist or nurse. Most participants expected the nurse to take greater responsibility for resolving the situation, despite the receptionist being the source of error. Irritation emerged as

the dominant emotional response in both conditions. However, the participants were more accepting of errors made by the robot.

To conclude, we can say that the responsibility spills over to co-present humans and the larger organizational system when the receptionist is a robot, while if the receptionist is human, they are held accountable as an individual and are expected to take responsibility for their actions. In addition, we found that participants strongly identified with the patient. We therefore suggest that it would be beneficial to study a similar scenario from the perspective of healthcare staff. Lastly, since the presented results stem from an exploratory pilot study, we recommend replicating the study with a larger sample and using both quantitative measures to assess attributed responsibility and qualitative measures to explore emotional responses.

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Dopaminergic Modulation of Sensory Gating in the Mouse Orbitofrontal Cortex: Toward a Developmental Account of Schizophrenia Vulnerability

by

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Sensory gating (SG), which is the brain's ability to suppress predictable or redundant sensory inputs, is a fundamental cognitive mechanism which underpins the mechanisms of attentional filtering and predictive processing (Adler et al., 1982; Micoulaud-Franchi et al., 2015; Sánchez-Morla et al., 2008). Reduced SG is one of the most robust and heritable endophenotypes of schizophrenia (SZ) and has been linked to perceptual flooding and to positive symptoms such as hallucinations and delusions (Luck & Gold, 2008; Sánchez-Morla et al., 2008). However, the cortical microcircuit mechanisms underlying SG, and how they mature, remain poorly understood.

In mice, SG is present in the orbitofrontal cortex (OFC) but appears absent in primary auditory cortex, suggesting that it relies on a local inhibitory microcircuit (Tripathi et al., 2021). More recent work implicates somatostatin-positive (SST) interneurons and dopaminergic modulation in OFC gating (Tripathi et al., 2025).

This work had two aims: to further examine the contribution of dopamine to SG in the OFC, and to prepare an experimental platform for later optogenetic experiments. We performed stereotaxic surgeries, viral injections, and local field potential (LFP) recordings in the OFC of mice, followed by histology and analysis. Auditory stimuli evoked strong responses in the OFC. In a paired-pulse paradigm, where two identical sounds are presented back-to-back, the response to the second sound was attenuated compared to the first, confirming SG in the region. Local injection of a dopamine D1-Recepor antagonist (SCH23390; n = 5) led to an impairment of this attenuation, while vehicle infusion (n = 4) did not. Furthermore, channelrhodopsin-expressing virus was injected into the OFC of SST-Cre mice (n = 2) to restrict expression to SST interneurons. These injections constitute preparation for future experiments; no optogenetic manipulation has yet been performed.

Optogenetic manipulation of SST interneurons will be used to test whether they are necessary for SG. The timing between light and sound will be varied to separate the actual modulation of SG from photoelectric artifacts, and the interactions between dopamine and SST interneurons

will be characterized. We further hypothesize that SG matures late, consolidating in early adulthood, which would overlap with the peak onset window for SZ. Candidate contributors are the NR2B-to-NR2A subunit switch (Wang & Gao, 2009), maturation of the dopamine system and its modulation of cortical interneurons (Tseng & O'Donnell, 2007), and increasing inhibitory tone of SST interneurons. If SST-to-$GABA_B$ signaling limits this maturation, $GABA_B$ agonists might induce adult-like gating in adolescent animals (Tripathi et al., 2025). This remains to be tested.

This data is limited to a small mouse sample and a pharmacological manipulation, so the SZ-related implications are speculative. If the above hypotheses are supported, they may point to inhibitory and neuromodulatory mechanisms as candidate targets for studying early intervention.

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Physically Embodied AI for Waste Classification: Distributed Cognitive Labour at Recycling Centres

by

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Recycling centres are cognitively demanding workplaces. Workers sort visually ambiguous recyclable material through repetitive motor tasks under conditions that have been characterised as both physically and cognitively taxing (Emmatty & Panicker, 2021). The consequences of misclassification; contaminated recovery streams and lost recyclables make this a cognitive task with significant real world consequences (Rousta & Ekström, 2013). To address this, AI classifiers are proposed as tools to support or replace human sorting work (Satav et al., 2023). This raises a question for cognitive science: how should recycling centres be designed to distribute the cognitive labour to best account for the strengths and needs of human workers and machines as new AI systems are introduced?

The primary contribution of this study is theoretical, while relying on a proof-of-concept of a physically embodied AI prototype to empirically frame the call for a distributed cognition account of how AI can reshape cognitive work at recycling centres. The system fine-tunes a ResNet18 network pretrained on ImageNet on the TrashNet dataset (Yang & Thung, 2016), a standard benchmark for waste classification (Aral et al., 2018) containing 2527 images across six categories. The classifier's prediction is sent via serial communication to an Arduino Uno

controlling a servo motor, with each category mapped to a distinct angle, producing a physical sorting action for every decision.

The model reached 82% accuracy on the TrashNet test set, which was collected in the USA. To test generalisation, 17 real Swedish household waste items were collected and classified in Gothenburg. Accuracy dropped to 65%, but more importantly, accuracy varied by material: cardboard and paper transferred well (100%), while plastic (25%) and glass (0%) performed poorly. The benchmark confusion matrix had already foreshadowed this weakness: plastic and glass were frequently confused.

This pattern of cross-cultural variability points to a concern: cultural bias. TrashNet reflects American trash and American recycling categories; deploying such a system in Sweden, where packaging and categories differ, shows these assumptions as performance failures and highlights the need to test classifiers in the contexts where they will be deployed.

Furthermore, adding an AI classifier to a recycling centre does not simply improve sorting. It changes what kind of cognitive system that centre is. Following Hutchins (1995), a recycling centre can be understood as a distributed cognitive system that sorts waste, and the decisions emerge from the coordinated activity of workers, infrastructure, and procedures, not from any single agent within it. Therefore, introducing an AI classifier might also categorically change it: new kinds of decision arise, such as when workers should trust or override the classifier's output, and the cognitive task is partly reshaped rather than simply reallocated. Our results highlight differences in classifier reliability across material categories, suggesting that sorting cannot be handed over to the classifier uniformly. This emphasises the need to design recycling centre workflows around the capacities of each agent and to consider the interfaces through which human workers, machines, and procedures coordinate.

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Can Personality Factors Derived from the Temperament and Character Inventory (TCI) Predict Cognitive Function in Older Adults?

by

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Key words: personality, neuroticism, cognitive function, episodic memory

Background: Personality is a relatively stable characteristic across the lifespan and may contribute to individual differences in cognitive aging. Previous studies have identified neuroticism as a potential personality risk factor for cognitive impairment; however, findings have been inconsistent. Furthermore, studies examining the psychometric properties of the Temperament and Character Inventory (TCI) in the Betula dataset have yielded mixed results.

Objective: This study aimed to improve understanding of individual differences in cognitive aging. Accordingly, we aimed to evaluate the psychometric properties and address inconsistencies in the TCI among middle-aged and older adults, and to examine whether the Neuroticism factor derived from the TCI predicts cognitive performance, particularly episodic memory, a cognitive function that is highly sensitive to age-related decline.

Methods: Participants were 1628 adults aged 40 to 86 from Wave 2 of the Betula study. Rasch analysis was used to assess item functioning. A sequential approach incorporating Exploratory Factor Analysis (EFA), Confirmatory Factor Analysis (CFA) and Rasch analyses for each retained factor was conducted to evaluate the TCI, identify well-functioning items, and

establish its underlying structure. Given prior evidence linking Neuroticism to cognitive decline, the Neuroticism factor was subsequently entered into a multiple linear regression analysis to predict episodic memory while controlling for age, sex, and education.

Results: Findings indicated acceptable model fit for EFA, CFA, and Rasch models. Six personality factors were identified: Neuroticism, Agreeableness, Emotional Stability, Cooperativeness, Self-Transcendence and Self-Restraint. Multiple linear regression analysis showed that higher Neuroticism was significantly associated with lower episodic memory performance ($B = -0.01$, $p < .001$), whereas higher education was associated with better performance ($B = 0.025$, $p < .001$). Higher age was significantly associated with poorer episodic memory ($B = -0.017$, $p < .001$). Episodic memory performance differed between men and women ($B = -0.13$, $p < .001$). The overall model explained 42.8% of the variance in episodic memory (adjusted $R^2 = 0.427$, $F(4,1607) = 300.70$, $p < .001$).

Conclusion: The findings suggest that Neuroticism may be an important personality-related correlate of cognitive ageing, particularly episodic memory performance, in middle-aged and older adults and highlight the potential utility of a psychometrically refined TCI for investigating individual differences in cognitive functioning across adulthood and older age.

When AI Shapes Memory: The Role of Warning Labels and Confidence Judgements

by

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The increase of artificial intelligence (AI) has raised concerns about its influence on human memory and the spread of misinformation. AI makes it possible to create realistic, manipulated images that are difficult to distinguish from authentic material, potentially increasing the formation of false memories (Xu et al., 2023; Pataranutaporn et al., 2025). Such content can become integrated into existing memory representations and is often experienced with high confidence (Loftus et al., 1978; Loftus, 2005). As a result, the misinformation can spread rapidly across social media platforms. Although platforms use warning labels on content to alert users to false or misleading content (Guo et al., 2023; Blank & Launay, 2014), it remains unclear whether these labels effectively protect users.

The present study therefore investigates whether AI-manipulated images increase false memories and whether subtle AI-labels can reduce these effects and influence confidence in memory judgements.

A between-subjects experimental design compared three conditions: a control group viewing original images, a group viewing AI-manipulated images, and a group viewing the same AI-manipulated images with a subtle, TikTok-inspired label. The sample consisted of 117 participants aged 19–29 years. Participants first viewed 24 original nature images during an encoding phase. After a distractor task, they viewed the 24 images again under one of the three experimental conditions. In the manipulated conditions, subtle visual details were added or removed. Participants then completed a recognition memory test featuring the 24 images again with the details masked, followed by confidence ratings on a 7-point scale.

One-way ANOVAs and Tukey's HSD post hoc tests showed a significant effect of condition on false memories, $F(2, 114) = 8.11$, $p < .001$, $\eta^2 = .12$. Participants exposed to AI-manipulated images reported significantly more false memories than participants in the control group. However, no significant difference was found between the manipulated condition with and without labels, suggesting that the subtle AI-label did not reduce false memories. A second ANOVA examining confidence ratings showed a significant overall effect of condition, $F(2, 114) = 4.49$, $p = .013$, $\eta^2 = .07$. However, the AI-label did not significantly reduce confidence compared to the AI-manipulated condition without a label.

Crucially, the control group exhibited a high baseline error rate. The visual complexity of the nature scenes likely forced participants into guessing. This makes it difficult to distinguish false memories, linked to source monitoring failures, from guessing. Consequently, group differences may reflect the control group's repetition advantage over a true AI effect. While this makes definitive conclusions uncertain, the results can still highlight memory's vulnerability in AI-shaped environments. If such an effect exists, realistic AI images may distort memory despite subtle warning labels. This creates false memories that people are highly confident in, further spreading misinformation. This illustrates how passive labels fail to provide the cognitive friction needed to disrupt automatic processing under bounded attention.

Future research should control task difficulty to isolate genuine reconstructive memory from guessing, determining if more salient and active labelling strategies can better protect users in digital environments.

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Microsaccades Track Feature-Based Content in Visual Working Memory

by

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Microsaccades are small fixational eye movements whose direction is biased toward covertly attended locations and toward locations maintained in visual working memory (Engbert & Kliegl, 2003; van Ede et al., 2019; de Vries & van Ede, 2024). Recent work further suggests that miniature gaze patterns can carry information about non-spatial visual features held in mind (Linde-Domingo & Spitzer, 2024). However, it remains unclear whether microsaccades also track non-spatial visual features when spatial location is held constant, and whether they reflect the selective maintenance and internal selection of task-relevant information. We addressed these questions by reanalyzing eye-tracking data from 12 participants performing a sequential orientation-recall task (Liljefors et al., 2024). Four oriented bars were presented sequentially at the center of the screen. At the start of each trial, a pre-cue indicated whether all four orientations should be remembered (load 4) or whether bars 2 and 3 were distractors that could be ignored (load 2). Each bar was followed by a short blank delay, and at the end of each trial, a retro-cue indicated which relevant orientation should be reported. During delay intervals, microsaccade alignment was defined relative to the preceding bar orientation, and during the retro-cue interval relative to the cued orientation. Microsaccades within ±45° of the reference orientation were classified as aligned. Microsaccade counts were analyzed using negative-binomial generalized linear mixed-effects models. During the blank delay intervals, microsaccade direction tracked task-relevant orientation information in the absence of visual input. For delays following bars 2 and 3, microsaccades were more often aligned than misaligned when these bars had to be maintained in load-4 trials ($p < .001$), whereas no alignment effect was detected when the same bars were distractors in load-2 trials ($p = .212$). For delays following bars 1 and 4, which were relevant in both load conditions, aligned microsaccades were more frequent in both load-4 and load-2 trials (both $p < .001$). Following the retro-cue, microsaccades again aligned with the orientation selected for report ($p < .001$), indicating internal selection of a stored VWM representation. This target-alignment effect was evident on high-accuracy trials ($p < .001$) but was not detected on low-accuracy trials ($p = .209$). Importantly, on low-accuracy trials, microsaccades instead aligned with the orientation that participants subsequently reported ($p = .009$), suggesting that microsaccade direction reflected the memory representation that ultimately guided behavior. These findings extend previous work linking microsaccades to spatial attention and location-based VWM by showing that microsaccade direction can also track selectively maintained and internally selected non-spatial visual features, even when all stimuli are presented at the same foveal location.

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When the Mind Speaks: How Does Anxious-Depression Relate to Metacognitive Outcomes Across Social-Emotional and Cognitive Domains?

by

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Background: Metacognition, the ability to evaluate one's own cognitive processes, plays a central role in adaptive behavior. Anxious-depressive (AD) traits are associated with alterations in metacognitive judgments, particularly reduced confidence, yet it remains unclear whether this alteration is domain-general, meaning it occurs consistently across different types of tasks (e.g., perceptual and social-emotional), or domain-specific, meaning it occurs only in certain types of tasks. Previous research is mostly focused on cognitive domains, paying less attention to the social-emotional domain despite evidence that anxiety and depression are associated with alterations in emotional processing. This study examined how AD relates to metacognitive bias (overall confidence), metacognitive sensitivity (the ability to accurately differentiate between one's correct and incorrect answers), and task performance across both perceptual and social-emotional trials.

Method: Data from a sub-clinical sample recruited online ($N = 418$; Mage = 31.60, $SD = 9.04$) were analyzed. Participants completed a decision-making task comprising two alternating trial types, each with 36 trials (72 trials in total). In one type, the dot-estimation trials, participants estimated the number of dots presented. In the other, the facial emotion recognition trials, designed using stimuli from the Emotion Recognition Task (Montagne et al., 2007), participants identified the facial emotion displayed. After each response, participants rated their confidence from 1 (guessing) to 6 (very sure). The AD dimension was derived using factor analysis of validated self-report psychiatric questionnaires. Accuracy, metacognitive bias (mean confidence across trials), and metacognitive sensitivity (area under the type-2 ROC curve; AUROC2) were examined as primary outcomes. Mixed-effects models tested associations between AD scores and each outcome, with a Task × AD interaction term (reflecting trial type) to assess domain specificity.

Results: Higher AD scores were associated with significantly lower confidence across both the dot-estimation and facial emotion recognition trials ($B = −0.09$, $p = .005$), with no significant Task × AD interaction ($B = 0.03$, $p = .084$), indicating that lower confidence was present in both trials. Higher AD scores predicted lower accuracy in the dot-estimation trials ($OR = 0.94$, $p = .008$) but higher accuracy in the facial emotion recognition trials ($OR = 1.11$, $p < .001$). Metacognitive sensitivity was not associated with AD traits in either trial type ($B = 0.006$, $p = .236$).

Conclusion: This study shows that anxious-depressive traits are associated with lower general confidence. In contrast, objective task performance varied by domain. Higher AD scores were associated with lower accuracy on dot-estimation trials but higher accuracy on facial emotion recognition trials. The better accuracy on facial emotion recognition trials is consistent with

heightened sensitivity to emotionally relevant information. Higher AD scores were not associated with metacognitive sensitivity in either trial type, suggesting that trial-level discrimination between correct and incorrect responses was unrelated to AD. Together, these findings suggest that metacognitive bias, but not metacognitive sensitivity, may represent a domain-general feature of anxious-depression traits. Future research could examine the social implications of this lower confidence and its potential as a treatment target.

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Mimadeck simulation platform: The development of a new software platform that will aid in constructing simulated environments for virtual agents

by

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Main goals

Virtual agents, embodied in a simulated environment, can be used to conduct experiments and test cognitive hypotheses (Oved et al., 2024). However, such simulation-based experiments are often constrained by the lack of suitable platforms that offer both realistic simulations, easy to share and replicate, and integration of AI techniques and methodologies to augment agent behavior. There is a general lack of software platforms with an intuitive user interface that makes it easy to fine-tune experiment conditions. Existing platforms frequently suffer from limited interoperability, inadequate integration with modern machine learning frameworks, and insufficient realism in terms of dynamic and physically accurate simulations.

The goal of the Mimadeck project is to address these limitations by developing an accessible and powerful software platform that integrates AI-techniques such as Reinforcement Learning (RL) and Large Language Models (LLMs) in an easy-to-use software package designed for

easy experimentation with virtual agents in a simulated environment. This platform would facilitate research primarily in cognitive science, but would potentially also be useful in related areas such as neuroscience and psychology (Eberlen et al., 2017). Additionally, it may also find application in fields such as robotics and embodied AI.

In essence, Mimadeck will allow for easy creation and maintenance of simulated environments that can be used to train virtual agents and test hypotheses by experimenting with various configurations of both agents and environments.

Project

Mimadeck will be designed to support innovative research by facilitating experimentation with virtual agents in simulated environments. To enhance the behavioral repertoire for the virtual agents, it will incorporate machine learning techniques such as Reinforcement Learning (RL) and even introduce the potential to enhance these algorithms with Large Language Models (LLMs). Furthermore, Mimadeck's modular design and user-friendly interface will allow researchers to easily fine-tune agent parameters and modify environments with readily exchangeable components. This will significantly reduce the barriers to conduct complex experiments and test hypotheses in silico.

Mimadeck will adopt a modular client–server architecture to ensure scalability and maintainability. The client component will be implemented as a cross-platform desktop application designed to run efficiently on all platforms, and will be distributed as open-source software to ensure broad adoption within the research community. The server component will be deployed on cloud infrastructure to provide robust computational resources capable of supporting reinforcement learning algorithms and interfacing with LLM services.

The client application will be developed using the Godot game engine since it offers several advantages particularly well-suited for academic and research settings: it is entirely free and open source under the permissive MIT license, supports high-performance rendering, and enables rapid prototyping through its flexible and intuitive scripting environment.

On the server side, the platform will be built using Python and FastAPI, enabling seamless integration with machine learning libraries and allow for easy interfacing with LLM services. This architecture will support both real-time and asynchronous interaction between agents and their environments.

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